

From plan to plot: designing for long-term crisis response in community gardens in Holyoke,
Massachusetts

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ABSTRACT

When federal food support for a city is threatened, why do some community gardens become vital sites of crisis response while others fall into neglect? This thesis investigates this question through a diagnostic analysis of seven community gardens in Holyoke, Massachusetts. Using a mixed-methods approach structured around Importance-Performance Analysis, I utilize municipal planning documents, site observations, and semi-structured interviews. I find that the formal "importance" ascribed to garden features is institutionally fragmented between city-led revitalization logic and housing authority amenity logic. This fragmentation manifests in differences between the planned importance and lived performance of a community garden. Gardens thrive where community-driven political action secures space for deep cultural practice, persist but remain limited where spatial security is achieved without collective infrastructure, and fail where agency is outsourced to institutional gatekeepers. I argue that crisis does not create resilience; it reveals what has been cultivated over time through legible boundaries, multifunctional infrastructure, and complementary modes of commoning. This challenges three homogenizing assumptions in planning literature: that gardens are inherently grassroots, that crisis can create resilient gardens, and that land tenure secures a garden's survival. The thesis concludes with three design principles for cultivating latent capacity in community food infrastructure, contributing to planning literature by operationalizing latent capacity as a diagnostic framework and to architectural studies by specifying how design decisions shape social outcomes under stress.

ACKNOWLEDGEMENTS	2
ABSTRACT	3
PREFACE	5
CHAPTER ONE: INTRODUCTION	6
A. Methodology	7
B. Contributions	11
CHAPTER TWO: LITERATURE REVIEW	13
A. Historic Overview of Community Garden Planning	13
B. Types of Agency in Community Gardens	16
C. Competing Frameworks of Food Security & Food Sovereignty	20
D. Takeaways	23
CHAPTER THREE: STUDY CONTEXT	24
A. Holyoke	24
B. Site Histories	28
CHAPTER FOUR: PLANNING & SPATIAL ANALYSIS	31
CHAPTER FIVE: SITE OBSERVATIONS	51
A. Nuestras Raices	51
B. Comparative Cases	63
1. Gardening the Community	63
2. Holyoke Food & Equity Collective (Holyoke Collective)	73
CHAPTER SIX: DIAGNOSING LATENT CAPACITY	75
A. Garden Classifications	75
B. Comparative Tensions	79
C. Implications for Crisis Response	83
1. Principle 1: Design for legible boundaries of responsibility	83
2. Principle 2: Design against micro-territories by designing for multifunctionality & mobility	86
3. Principle 3: Design for complementary modes of commoning.	89
CHAPTER SEVEN: CONCLUSION	91
List of Tables and Figures	97
References	99

PREFACE

What happens when the federal food support for a city suddenly vanishes? I first asked this question in the fall of 2025, when for the first time in US history, the Supplemental Nutrition Assistance Program (SNAP) could go unfunded due to a government shutdown and the Trump Administration's refusal to release SNAP contingency funds. In Holyoke, Massachusetts—one of the most food-insecure cities in the state—the cuts landed with particular force. As Mayor Joshua Garcia noted, SNAP recipients in Holyoke are "overwhelmingly made up of senior citizens and children," making the sudden loss of benefits especially devastating for the city's most vulnerable (Loughrey & Gardyna, 2025; Lisowski & Gardyna, 2025; Munford, 2025).

I first encountered Holyoke through its reputation. I came expecting to study a singular model of resistance, to draw lessons from landmark cases like Esperanza Garden in New York and D-Town Farm in Detroit, and to trace how communities defend their gardening plots during times of crisis. But as I spent more time in Holyoke, I realized that the lessons of Esperanza and D-Town could only take me so far. Those gardens had been shaped by different histories, different political landscapes, different configurations of land and power. They could not tell me why, in Holyoke, some gardens thrived while others were severely neglected within just a two-mile radius. In hyperlocalizing my study to Holyoke, the result is an analysis of seven gardens within a single city, a single organization, and two planning regimes – because if crisis resilience is built locally, then the scope of analysis should be local.

CHAPTER ONE: INTRODUCTION

Infrastructure gaps often remain marginal until crises like natural disasters, pandemics, or system failures force them into the limelight. A heatwave exposes an aging power grid unable to meet demand. A bridge collapse reveals decades of deferred maintenance. A winter storm cripples a water system never designed for extreme weather. These moments of failure expose what was always there: the vulnerabilities embedded in systems that functioned adequately under normal conditions but lacked the capacity to withstand stress.

Community gardens are infrastructure too –social infrastructure, ecological infrastructure, and food infrastructure –yet they are selectively treated as such. They occupy an ambiguous position in the American urban landscape where they are simultaneously grassroots spaces and institutional projects, temporary land uses and permanent community anchors, and sites of individual food production and collective cultural practice. When the COVID-19 pandemic exposed deep vulnerabilities in food supply chains, and when unprecedented cuts to the SNAP abruptly removed a critical layer of nutritional support for millions, some community gardens became central locations for community response. Others, seemingly well-located and well-equipped, faced declining stewardship. This divergence reveals the latent resilience or latent vulnerability embedded in a garden's design, governance, and daily life.

This thesis investigates why some community gardens become vital during crises while others do not. Rather than treating crisis as the central subject, I use it as a litmus test that exposes the gaps between how gardens are planned and how they are maintained, between the importance ascribed to certain features in institutional documents and the performance of those features in practice, especially under stress. A garden's performance under stress reveals whether its agency is held collectively or outsourced to institutions, whether its social infrastructure enables mutual aid or merely individual production, and whether its cultural practices are living and shared or static and proprietary.

The historic literature on community gardens reflects a persistent planning culture that treats gardens as interim uses rather than permanent public resources. This pattern reveals two systemic failures: (1) the gap between crisis-driven garden creation and post-crisis stewardship, and (2) the gap between user-initiated spaces and spaces created from institutional partnerships.

The myths that structure garden planning would have us believe that new gardens will spring up in response, that crisis creates resilience. What I find in response to the question "what happens when the federal food support for a city suddenly vanishes" is actually the following. I argue that the gardens that thrive under stress are those whose latent capacity has been cultivated over time, through design that makes boundaries legible, infrastructure that enables collective practice, and governance that distributes care rather than outsourcing it to institutional gatekeepers.

A. Methodology

This study's use of the term gap follows directly from the literature review, which identifies two systemic failures in garden planning. The first is the gap between crisis-driven garden creation and post-crisis stewardship. The second is the gap between user-initiated spaces and their integration into long-term planning. Throughout this thesis, the gap refers to the divergence between what planning documents assume gardens need and what actually enables them to persist under stress.

To examine these gaps, this study studies seven gardens across Holyoke, Massachusetts, using a mixed-methods design structured around an Importance-Performance Analysis (IPA), inspired by studies evaluating marketing and sustainability interventions (Martilla & James, 1977; Deng & Pierskalla, 2018). The IPA framework systematically identifies differences between planned intent, or "importance", and lived reality, or "performance" – a divergence central to understanding gardens' adaptive capacity. This research expands typical IPA analysis from abstract user expectations to the material and social realities of existing garden sites.

1. Document and Spatial Analysis

This analysis reconstructs the formal, designed "importance" of garden features by examining institutional and planning documents. Data sources include spatial plans, municipal planning documents, organizational strategic plans, and archival literature from organizational histories, news articles, and prior studies. Content analysis generates a taxonomy of important features for each garden, organized into spatial attributes like plot layout and infrastructure, political attributes like governance rules, and cultural attributes like plant selection and educational content.

2. Field Observations

To document on-ground "performance," I conducted structured site visits from October to November 2025 employing condition assessments, photographic surveys, and spatial analysis of "cues to care" – material traces such as plot maintenance, informal social infrastructure, crop diversity, and signs of neglect that serve as proxies for user engagement, governance models, and cultural practice. Rather than traditional real-time behavioral mapping (Ittelson et al., 1970), this study infers patterns of use, stewardship, and social meaning from cumulative spatial adaptations visible at each site (Goličnik, 2005).

Each garden was classified into one of three performance states. "Lively" signifies high adaptive social use and evidence of collective life beyond individual cultivation. This evidence spans communal cooking/gathering spaces, user-built structures, cultural markers, and active programming infrastructure. "Alive" signifies functional but primarily individual or family use sites; this means maintained plots and basic infrastructure that may lack communal spaces or evidence of collective gathering. Finally, "Neglected" signifies underutilized, overgrown plots, minimal recent cultivation, deteriorating infrastructure, and an absence of user-built adaptations

My positionality also matters here: I engaged these spaces as a researcher from outside the community, documenting physical and material conditions as trace evidence. It was only after initial site visits that I conducted in-depth interviews with Nuestras Raíces staff, coordinators, and growers. Their generosity revealed the immense, often invisible labor sustaining these gardens – context a purely spatial analysis cannot capture. The performance assessment that follows is therefore a diagnostic mapping of material outcomes instead of a critique of the organization's value or effort.

3. Importance-Performance Analysis and Latent Capacity

This analytical synthesis brings together the document analysis and field observations to identify differences and assess each garden's latent capacity. Borrowing from disaster risk reduction scholarship, latent capacity is defined here as the functional resources, spatial adaptations, and social networks embedded within a garden that, while not explicitly designed for crisis response, can be activated or repurposed during systemic shocks such as sudden food assistance cuts (Manyena, 2006).

4. Site Selection: Nuestras Raíces

Nuestras Raíces, founded in 1992 by Puerto Rican migrant farmers, represents a culturally rooted, institutionally embedded model. Its decade-spanning presence in Holyoke responds directly to the city's deindustrialization, the concentration of Puerto Rican communities in marginalized neighborhoods, and resulting spatial inequities in food access (Finnell et al., 2022; Lao-Montes, 2007). The organization's gardens are an appropriate case for three reasons.

First, the garden network spans two distinct planning regimes: the urban core (governed by municipal redevelopment frameworks) and West Holyoke (governed by Holyoke Housing Authority management logic). This institutional variation provides a natural experiment for examining how different planning contexts shape garden outcomes.

Second, Nuestras Raíces's thirty-year history allows examination of latent capacity's durability across time. The organization has weathered cycles of partnership and disinvestment, pandemic disruption, and coordinator turnover. Selecting gardens that reflect this temporal depth – from foundational La Finquita (1992) to more recent sites like El Renacer (2009) – reveals which conditions sustain gardens and which allow decline.

Third, Nuestras Raíces embodies the tension central to my agency framework: gardens as sites where top-down institutional agendas meet bottom-up, user-initiated practice (Staeheli, Mitchell & Gibson, 2002; Pudup, 2008; Saugeres, 2002). Its gardens range from those tightly integrated with municipal and housing authority partnerships (Falcetti Towers, Hidden Treasure, El Renacer) to those where community-driven political action secured space and continues to shape use (Cuenta Conmigo). This variation tests the hypothesis emerging from my literature review: gardens cultivating integrated spatial, political, and cultural agency will exhibit greater latent capacity for crisis response than gardens where agency is individualized or outsourced to external institutions.

To operationalize this comparative case study design, I selected seven Nuestras Raíces gardens that vary systematically along three axes: (1) spatial configuration and urban context, (2) governance model and institutional partnerships, and (3) the presence or absence of communal infrastructure enabling collective agency. I do not claim these seven gardens capture the full diversity of community gardens across Holyoke or all Puerto Rican-led gardening initiatives in

the diaspora. Rather, the selection functions as a microcosm of key garden models emerging from the literature and my study context.

5. Interviews

I conducted semi-structured interviews with the following Nuestras Raíces stakeholders: a Community Garden Coordinator, a Director involved in organizational partnerships, and a grower at the urban farm La Finca. Questions were modified based on each interviewee's personal history to promote self-reflection and dialogue around the meaning and politics of community gardens. These interviews are not representative of the broader Nuestras Raíces collective; they function as grounded expertise from individuals whose organizational roles and lived experience grant deep, situated knowledge of the gardens' histories, governance, and daily realities. Their testimony provides a causal narrative that spatial analysis alone cannot supply. To protect privacy and maintain trust, I refer to them throughout by their organizational roles rather than their names.

6. Comparative Cases

To test whether the patterns observed in Nuestras Raíces generalize across different organizational logics, I selected two additional cases representing alternative models of urban food resilience. Holyoke Collective embodies a practice-based, landless mutual aid model that stands in contrast to Nuestras Raíces's territory-based approach. Gardening the Community represents an infrastructurally-focused, youth-led hub-and-spoke model. Together, these three cases allow this study to probe whether latent capacity for crisis response is better cultivated through long-term cultural integration (Nuestras Raíces), mutual aid networks (Holyoke Collective), or infrastructural investment (GtC).

I conducted a semi-structured interview with a founding member of Holyoke Food & Equity Collective (Holyoke Collective), following a similar format to Nuestras Raíces but tailored to the Collective's distinct structure and history. Ideally, I would have conducted a parallel interview with a Gardening the Community (GtC) stakeholder. However, due to time constraints and inability to secure interviews, I instead relied on secondary sources and site visits to Hancock Street Farm to observe spatial differences firsthand. While this limits organizational

and experiential data for this case, the spatial evidence and documentary record nonetheless offer valuable points of contrast.

This method operationalizes the action anthropology principles guiding this research (Taplin et al., 2002). The recursive design of using field observations informing interviews, patterns prompting revisits, hypotheses tested against new evidence is aimed at preventing premature conclusions. The IPA framework diagnoses differences between planned importance and lived performance without claiming to fully know each garden's needs. To this end, classifications of "Lively," "Alive," and "Neglected" are diagnostic tools. The community retains the right to interpret, reject, or act on these findings. The method also offers a replicable framework a garden could adapt toward its own self-determined goals.

B. Contributions

This thesis makes three core contributions. Firstly, I provide a diagnostic framework for evaluating the latent resilience of community gardens. While existing planning literature has documented the proliferation of community gardens and their benefits, it has not offered replicable methods for assessing why some gardens thrive under stress while others fail. Secondly, I challenge the homogenizing assumptions that have structured garden planning and scholarship for over a century. By tracing the material traces of seven gardens across a single organization and comparing territory-based and practice-based models of commoning, I reveal that resilience is an emergent outcome (not a property) of specific design, governance, and maintenance choices. Lastly, I propose three principles for designing gardens capable of weathering crisis, each emerging from the tensions revealed in Holyoke: design for legible boundaries of responsibility, design against micro-territories by designing for multifunctionality and mobility, and design for complementary modes of commoning through shared territories *and* shared practices. These principles contribute to planning literature by offering actionable criteria for municipal and institutional partners investing in community gardens, and to architectural studies by specifying how spatial configuration, planting scale, and infrastructural decisions shape social outcomes under stress.

This thesis proceeds in seven chapters that move from problem to context to diagnosis to design recommendations. Chapter Two reviews the literature on community garden history,

exposing three homogenizing assumptions – namely, that gardens are grassroots, that crisis creates gardens, and that land tenure secures survival – that this thesis challenges. Chapter Three grounds the study in Holyoke, tracing Puerto Rican migration, the farming revolution, and bicultural practices. This context is essential because Holyoke's gardens cannot be separated from the people who built them. Chapter Four analyzes Holyoke's planning documents and site observations, revealing that "importance" is institutionally fragmented between revitalization logic and amenity logic. Chapter Five compares Nuestras Raíces with two alternative models, Holyoke Collective which is practice-based and Gardening the Community which is infrastructure-based, revealing that resilience involves trade-offs between territory and practice, and permanence and flexibility. Chapter Six synthesizes these findings to answer why some gardens become vital during crisis, extracting three design principles: legible boundaries, multifunctionality and mobility, and complementary modes of commoning. Finally, Chapter Seven concludes by arguing that crisis does not help create resilience; it reveals what has been cultivated over decades through planning, design, and governance.

CHAPTER TWO: LITERATURE REVIEW

The literature on community gardens is, in many ways, a literature of mythmaking. Planners and scholars have long treated gardens as inherently grassroots, as community-driven interventions, and as spaces whose survival hinges on mainly land tenure. Community gardens usually develop on vacant or underutilized urban land as places where participants cultivate and maintain vegetation. While some gardens are laid out strictly for the utilitarian distribution of gardening plots, others can serve as de facto parks with lawns, picnic areas, and play equipment. Through the provision of gardening space, participants not only enjoy the opportunity to grow plants but often voice satisfaction related to other personal and social needs (Kaplan, 1973). This enjoyment often extends outward to encourage others through demonstration areas and educational programs that teach about gardening, ecology, and cultural traditions. The problem is that these celebratory descriptions obscure the systemic failures that emerge when eras of crisis or joy pass and institutional attention shifts. To examine those instances, I review the historical arc of garden planning in the U.S., the types of agency gardens cultivate, and the competing frameworks of food security and food sovereignty.

The goal of this chapter is not merely to summarize existing research on community gardens, but to show how the planning literature has built three homogenizing assumptions that this thesis will challenge. First, the assumption that gardens are inherently grassroots and emerge spontaneously from community action. Second, the assumption that crisis creates gardens—that gardens spring up in response to need and serve as effective relief mechanisms. Third, the assumption that land tenure secures garden survival. This forms the basis for understanding how Holyoke challenges these assumptions.

A. Historic Overview of Community Garden Planning

The process from planning to maintenance has been well-documented in community garden literature. Beyond showing a trend in the proliferation of community gardens, the following overview provides the normative underpinnings of the garden planning profession. With the advent of advocacy planning in the 1960s and 70s, planners recognized community gardens as participatory assets without necessarily addressing the implications for land use and open space planning. In the process, community gardens have come to represent a model of

user-initiated open space without attention to their long-term potential in city planning. Ambiguity in community garden planning as a "public good" raises important questions regarding civic and local responsibilities for maintaining that public space – a question central to my focus on maintenance under stress.

The intermittent public support for gardens in the US is shown by the range of garden archetypes, such as the garden as an alternative to charity in the 1890s and the garden as a tool for beautification nowadays. Notably, gardens have been repeatedly mobilized in response to social, economic, and environmental crises, especially acting as a tool for food security. During the depression of 1893-1897, gardens were promoted as an alternative to direct charity. Detroit's "Pingree Potato Patches" served as a model, providing plots to unemployed families on temporarily donated land. However, once industrial employment rebounded, most projects were discontinued without plans for ongoing maintenance or land preservation (Cialdella, 2014). In the 1890s, school gardens also emerged as tools for teaching ecology, work ethic, and American values to children, especially immigrants. Supported by civic groups and later the US Bureau of Education, these gardens were rarely permanent and often relied on borrowed land (Trelstad, 1997).

World War I spurred a national gardening campaign to supplement domestic food supplies and support allies. Federal agencies and volunteer groups promoted "war gardens" as a patriotic duty; to get enough food to win, the Allied peoples had to utilize every possible garden spot, including backyard areas and vacant town lots. Post-war, despite advocates encouraging the continued support of home and community gardens, institutional support from public agencies quickly dissolved (Pack, 1919). Gardens reappeared as relief projects during the Great Depression, providing food and occupation for the unemployed. Programs were run by federal, state, and philanthropic entities but were again seen as temporary – phased out as New Deal public works programs began. For example, federal funding was withdrawn in 1937 with the advent of the food stamp program (Colcord & Johnston, 1933). Although food security was less critical than in WWI, victory gardens were promoted for morale, recreation, and civic beautification. In addition to their cost-saving benefits, a well-cited purpose of theirs was to provide an opportunity for gardening by urban dwellers and others who lack suitable home

garden facilities (National Agricultural Library, 1942). Federal and local committees provided guidance, but once again, most gardens were abandoned post-war (Elsworth, 1948).

The 1970s saw a resurgence of community gardens driven by grassroots activism, environmental awareness, and responses to urban disinvestment. Organizations like the American Community Gardening Association (1978) emerged, yet land tenure remained insecure, and gardens were rarely included in long-range open-space planning. Continuing into the 2000s, even successful gardens face persistent threats of eviction when property values rise, reflecting a planning culture that continues to treat gardens as interim uses rather than permanent public resources (Gralińska-Toborek, 2021).

Several homogenizing assumptions emerge from this historical pattern. First, there is the assumption that gardens are inherently grassroots; they emerge spontaneously from community action rather than being shaped by institutional agendas. Yet the historical record reveals that gardens have been repeatedly mobilized by federal, state, and municipal programs to serve larger social, economic, and environmental objectives, making them "laboratories for a particular agenda" as often as sites of grassroots resistance. Second, there is the assumption that crisis creates community gardens; gardens spring up in response to need and serve as effective relief mechanisms. Yet the pattern of crisis-driven creation followed by post-crisis abandonment suggests that rather than creating durable garden infrastructure, crisis creates temporary projects whose stewardship dissolves when institutional attention shifts. Third, there is the assumption that land tenure secures garden survival. Yet the historical literature documents gardens with secure tenure that nonetheless declined, and gardens on borrowed land that persisted through political mobilization.

This competition between these assumptions made by planners and the reality of how the gardens are maintained reveals two systemic gaps: (1) the gap between crisis-driven garden creation and post-crisis garden stewardship, and (2) the gap between user-initiated garden spaces and their integration in long-term planning. These gaps are features of a planning culture that this thesis seeks to address.

B. Types of Agency in Community Gardens

The historical pattern of garden planning establishes the interesting narrative that gardens are sites where top-down institutional agendas meet bottom-up, user-initiated practice. This tension plays out directly in the types of agency that gardens and urban farms are designed to foster. Moving beyond the simple provision of space, we can identify three distinct, though often overlapping, types of agency cultivated in these environments: spatial, political, and cultural. Furthermore, these types of agency operate on a spectrum from individual, manifesting in a personal plot and a private harvest, to collective, manifesting in communal areas, shared decision-making, and cooperative distribution.

Spatial agency represents the most fundamental and visible form of power exercised through community gardens: the capacity to transform, occupy, and claim control over underutilized urban land – a physical reclamation of space. However, as history warns, this agency is perpetually precarious without secure land tenure, often dissolving post-crisis when institutional priorities shift. The historical arc of New York's community garden movement, a well-documented case study, exemplifies this agency in its physical transformation, political claim-making, and perpetual defense.

Gardens often emerge from the "borrowed landscapes" of urban decay in vacant lots filled with rubble, garbage, and the signs of disinvestment (Schmelzkopf, 1995; Staeheli et al., 2002). The act of clearing these spaces, building soil, planting, and erecting fences is a material assertion of presence and care, transforming "junk-laden spaces" into productive, safe, and socially vibrant commons (Schmelzkopf, 1995, p. 371). This transformation creates use-value such as non-commodified benefits like recreation, community, and safety, directly challenging the prioritization of a plot's exchange-value in the real estate market (Eizenberg, 2012; Harvey, 1989). This material act is inseparable from a political claim to the "right to the city" (Lefebvre, 1996). As Staeheli, Mitchell, and Gibson (2002) argue, the gardens of New York became sites of struggle between two competing rights frameworks: the city's formal property rights (its legal entitlement to sell land for development) and the gardeners' communal rights forged through "sweat equity" arising from cultivating the land and the ongoing production of community space. Gardeners' claims were not based on legal title but on use, need, and the social value they

created, asserting a right to shape their environment according to communal rather than commercial logic.

However, this spatial agency is inherently precarious, existing in a state of perpetual contestation. As Schmelzkopf (1995) details, gardens are often established as "interim uses" on land held for future development. Their existence is secured only through continuous political mobilization and defense. The landmark battles in New York during the Giuliani administration, where gardens were threatened with auction, reveal how spatial agency must evolve from creation to protection. This defense mobilizes what Eizenberg (2012), drawing on Henri Lefebvre, identifies as the triad of material space, lived space, and representations of space. Gardeners defended not just the physical plots (material space) but also the cultural meanings and social bonds they housed (lived space), while simultaneously producing new counter-knowledge about urban justice, community, and equitable development (representations of space) to legitimize their claims. The resolution to these conflicts often involves a paradoxical privatization to protect the public. The transfer of garden lands to private land trusts (like the Trust for Public Land) saved them from development but introduced a new layer of institutional control, potentially diluting the grassroots, autonomous character that defined their spatial agency (Eizenberg, 2012, p. 769). This outcome underscores that spatial agency is a continuous process of negotiation within shifting power structures. Ultimately, the community garden manifests as a commons where the collective right to urban land is performed, defended, and relentlessly renegotiated against the pressures of the neoliberal city.

Political agency emerges from the collective action required for this defense. It is the capacity for community organizing, coalition-building, and engaging with (or resisting) municipal planning processes. As seen in the earlier discussion of New York's garden movement, gardens were sites of explicit political resistance –bottom-up efforts to reclaim urban space from disinvestment and gentrification. However, Pudup (2008) argues that under "roll-out" neoliberalism, where the state attempts to manage the social consequences of marketization through paternalist interventions (like public-private partnerships), the political character of many contemporary garden projects has shifted (Pudup, 2008). While they may still use the language of resistance, projects like The Garden Project and The Edible Schoolyard are often orchestrated by non-state actors and framed around individual transformation (personal

responsibility, self-improvement, and moral reform) rather than collective mobilization against gaps in state provisions. This reflects a neoliberal political rationality that depoliticizes social issues, redirecting political agency inward toward the cultivation of "citizen-subjects" who adapt to systemic inequalities through personal change, rather than outward toward structural critique or collective action. Thus, political agency in these gardens exists on a spectrum: it can be a tool for community empowerment and resistance to marginalization, but it can also be co-opted into a technology of governance that prioritizes individual adaptation to political challenges.

This tension between collective and individualized political agency is not merely a theoretical concern; it has material consequences for how gardens function in times of crisis. Gardens that cultivate a collective political agency may develop the social cohesion and organizational capacity to respond adaptively to shocks, like becoming sites of mutual aid. In contrast, gardens oriented toward individualized adaptation may lack the shared purpose and collaborative infrastructure necessary for collective crisis response, rendering them more susceptible to neglect when institutional support falters. This gap between the political logic embedded in garden planning and those enacted in daily practice forms a critical lens through which to analyze why some gardens thrive as sites of community resilience.

In studying how urban agriculture is experienced in daily practice, a common theme in the literature is "counter-tradition" as expressed by Lefebvre (1991) where the garden can be seen as a "differential space," a social space of continuous negotiation and possibility. This resonates strongly with the agency framework where the garden is not only a provided space (spatial agency) or a base for political mobilization (political agency), but is also a vital ground for cultural practice and knowledge production. This cultural practice is localized and experimental (Leary-Owhin & McCarthy, 2019). Cultural agency is the most profound, yet most contested and frequently misunderstood dimension. Institutions often pay lip service to cultural relevance by narrowly defining it as "access to culturally relevant foods." This approach, however, risks being merely an ingredient replacement—swapping kale for collard greens or tomatoes for tomatillos within an otherwise standardized, industrial production and distribution system (Guthman, 2008). It treats culture as a consumable product.

The work of Saugeres (2002) provides a critical lens to understand why this simplistic approach fails and what is truly at stake. Saugeres' study of French farm families reveals that a deep, legitimate connection to agricultural land and practice is not just about what is grown, but how and by whom, based on embodied, inherited knowledge. The male farmers in her study legitimate their mastery through a discourse of being "born into" farming, having the land "in the blood," and possessing an innate, sympathetic understanding of nature (Saugeres, 2002, pp. 377-378). While Saugeres uses this to critique the patriarchal exclusion in French farms, this also crucially highlights that authentic cultural agency in agriculture is rooted in knowledge systems, relationality, and identity –not just crop selection. This expanded understanding of cultural agency directly informs the analysis of gardens and farms in immigrant and diasporic communities, such as those studied in the context of neoliberalism. It supports the argument that spaces like community plots are not just sources of ingredients but are vital for homeland re-creation, intergenerational knowledge transfer, and the assertion of a cultural identity that is actively practiced and negotiated through the processes of growing, harvesting, and sharing food (Zail, 2023). The struggle for cultural agency, therefore, is a struggle to define and control the entire socio-ecological context of food production, preserving the "how" and "why" alongside the "what."

Furthermore, cultural agency involves ritual and non-commercial exchange. The study of African American community gardens by White (2011) in "Sisters of the Soil" shows how gardens serve as spaces for spiritual nourishment, collective healing from trauma, and the affirmation of an agrarian heritage often erased by history. Similarly, Ramirez (2015) examines Latino immigrant gardens as hubs for fostering *confianza* (trust) and informal social support networks that are crucial for navigating life in a new country. Here, the act of sharing seeds, harvests, and meals enacts a cultural logic of reciprocity that stands in contrast to market-based transactions.

The danger of institutional co-option lies in severing these deep cultural practices from the physical space. When a garden project is designed with only food yield or nutritional metrics in mind, it can inadvertently displace cultural knowledge with technical expertise (Trauger, 2017). The gardener becomes a laborer following an extension agent's plan rather than a knowledge-holder steering the process. Therefore, assessing a garden's cultural agency requires

looking beyond planting lists to ask: Whose knowledge guides the growing process? Are there spaces for spontaneous social gathering and ritual? Does the garden allow for expressive landscapes –arrangements of plants, altars, or structures that reflect cultural values? The gap between planning and maintenance is often widest here: a garden may be planned with culturally familiar crops, but it also needs to be maintained in a way that fosters knowledge transmission and social meaning.

With these various manifestations of agency in mind, once again we see several homogenizing assumptions made about community gardens that deserve some scrutiny. First, there is the assumption that spatial agency, once secured, endures. Yet the history of New York's gardens reveals that spatial agency requires continuous defense since tenure without ongoing political mobilization can be lost. Second, there is the assumption that gardens are sites of collective political agency by default. Yet Pudup's critique reveals that many contemporary garden projects cultivate individualized adaptation rather than collective mobilization, redirecting political energy inward toward personal transformation rather than outward toward structural change. Third, there is the assumption that cultural agency can be designed through crop selection. Yet the deeper literature on cultural practice reveals that cultural agency is rooted in knowledge systems, relationality, and identity –in the "how" and "why" of growing. Crisis, I argue, exposes these assumptions like in the scenario where a garden whose cultural agency is reduced to crop selection can lose its meaning when the knowledge-holders depart, as Chapter Six will show.

C. Competing Frameworks of Food Security & Food Sovereignty

Alongside the proliferation of community gardening in US cities, a growing academic literature has largely celebrated its overall social and environmental benefits, such as community development, social capital, and improved health and well-being (Alaimo et al., 2010; Kingsley et al., 2006). Concurrently, as discussed above, an "activist-practitioner" narrative frames gardens as political and cultural spaces, linking them to broader goals of food security and resistance to neoliberal food systems (Colasanti et al., 2012; Horst et al., 2017). Within this context, the terminology of urban agriculture has gained prominence as a more ambitious and policy-oriented framework in an effort to scale these food security benefits, often subsuming

community gardens within its scope (Beilin & Hunter, 2011). This creates an ambivalence of what a community garden and its small-scale interventions should look and feel like.

The case study of Prairie Urban Farm (PUF) by Granzow and Jones (2020) vividly illustrates the multiple interpretations of urban farming. PUF was officially framed as an urban agriculture project focused on sustainability and food security, where practices include permaculture techniques, localized planting strategies, and collective management models. Despite this official description, volunteers' everyday understandings of PUF were diverse and ambivalent: some viewed it as a scientific site for permaculture experimentation, others emphasized productivity and efficiency, and many simply understood it as a community garden. This ambivalence highlights the importance of looking beyond singular discursive frames. The study argues that focusing solely on official narratives obscures the rich meanings embedded in everyday urban agriculture, including dimensions of encounter, commoning, escape, experimentation, and informality.

Therefore, the value of community gardens cannot be assessed solely through their official functional goals, such as food yield or sustainability metrics. Instead, they should be understood as a reflexive practice through which participants explore, understand, and respond to the challenges of a sustainable future. In this sense, the messiness and contradictions of gardens are not flaws but central features of their character as open-ended social experiments, preserving the potential to reimagine and rebuild urban life within shifting and uncertain contexts. This leads to a critical theoretical divide in how such spaces are understood between the dominant paradigm of food security and the more radical, process-oriented vision of food sovereignty.

Food security is institutionally defined by entities such as the USDA and the UN Food and Agriculture Organization (FAO) as existing "when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life" (FAO, 1996). While this definition is comprehensive, critical scholars argue it remains apolitical and ends-focused, emphasizing measurable outcomes like caloric intake, nutritional access, distribution efficiency (Patel, 2009; Jarosz, 2014). Under this framework, a community garden can be assessed as a food security

intervention based on its yield, nutritional output, or number of households served, aligning it with top-down policy objectives and institutional metrics.

Food sovereignty, in contrast, is an explicitly political and process-oriented concept. First articulated by the global peasant movement La Via Campesina in 1996, it asserts "the right of peoples to healthy and culturally appropriate food produced through ecologically sound and sustainable methods, and their right to define their own food and agriculture systems" (Via Campesina, 1996). Scholar-activist Raj Patel (2009) emphasizes its grounded, experiential nature: "Food sovereignty is the sort of thing one knows when one sees," a definition that emerges from grassroots struggle. This framework shifts the focus from access to control: who governs land, seeds, labor, knowledge, and cultural meaning? Consequently, its spatial and social requirements differ fundamentally from those of food security. A space cultivating food sovereignty must enable self-determination, cultural reproduction, collective governance, and ecological resilience; in other words, functions that are relational, dialogic, and embedded in practice (Trauger, 2017).

To what extent do these divergent logics materialize in urban community gardens? The question remains critically underexplored. This gap persists because food sovereignty is driven by the intersection of necessity, identity, and livelihood –a convergence that is deeply personal, culturally specific, and place-based (Alkon & Mares, 2012; White, 2018). It cannot be implemented as a standardized program but must be cultivated through the very "messy" practices of commoning, experimentation, and cultural negotiation observed by Granzow and Jones (2020).

A final set of homogenizing assumptions emerges from this theoretical divide. First, there is the assumption that productivity is the measure of a garden's value. The food security framework, with its emphasis on measurable outcomes like yield and nutritional output, tends to reduce gardens to their productive function, ignoring the social, cultural, and political dimensions that the food sovereignty framework prioritizes. Second, there is the assumption that permanence is the only path to resilience – an assumption similar to our discussion of securing land tenure. The food sovereignty framework, with its emphasis on land control and

self-determination, can obscure the possibility that distributed, practice-based models might achieve resilience through different means.

D. Takeaways

The various archetypes of community gardens have tended to treat gardens as monoliths. The literature reviewed here, which has structured garden planning and scholarship for over a century, builds myths surrounding grassroots origins, crisis-driven creation, the primacy of land tenure, the default collectivity of political agency, the reducibility of cultural agency to crop selection, and the necessity of permanence for resilience. Each of these assumptions, I have suggested, can be exposed by crisis, revealing whether a garden's agency is held collectively or outsourced to institutions.

As Langegger (2010) argues in his study of Denver's community gardens, the prevailing research framework often homogenizes these spaces, collapsing their complexity into a single category. Drawing on Foucault's concept of heterotopia (that is, spaces that are simultaneously real and mythic, capable of holding multiple, often incompatible meanings), Langegger instead proposes that community gardens should be understood as plural, tension-filled spaces that resist easy categorization. They exist in productive tension with the normative spaces of the city, offering alternate orderings of socio-spatial relations. This thesis addresses this gap by adapting Langegger's framework to a different question: not whether gardens are heterotopias, but how their heterotopic qualities shape their capacity to weather crises and sustain longer-term food distribution.

CHAPTER THREE: STUDY CONTEXT

A. Holyoke

Holyoke, Massachusetts, has the highest per capita concentration of Puerto Ricans of any city (with at least a population of 5000) in the US outside of Puerto Rico proper, cited as 45.8% in the 2020 US Census (US Census Bureau, 2023).

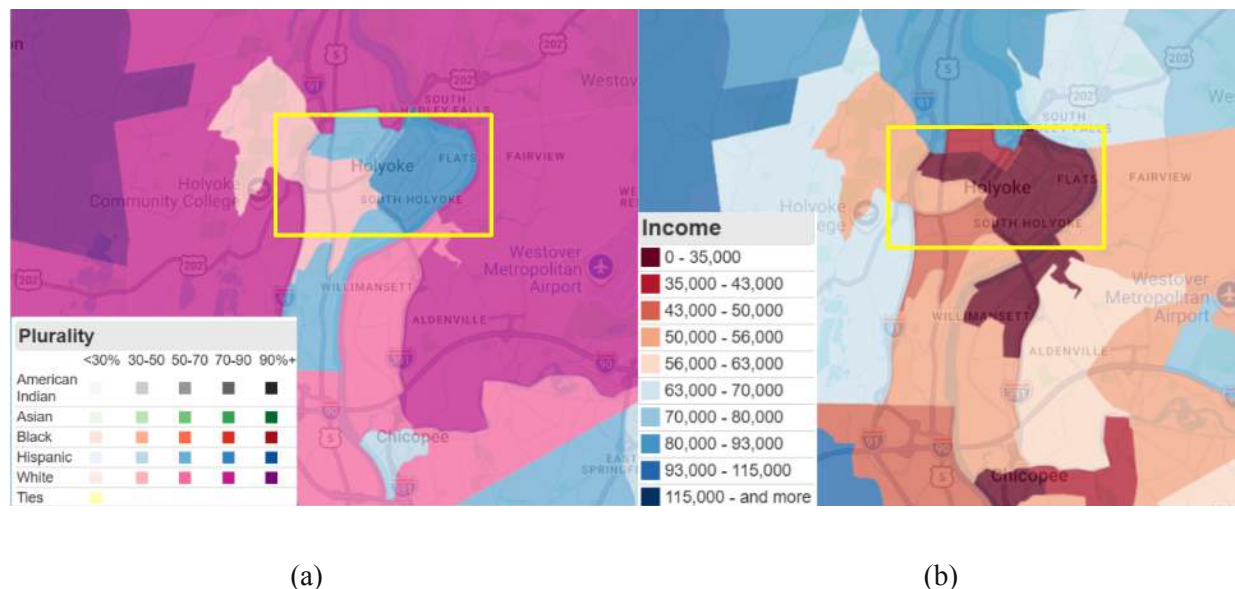


Figure 1: (a) tract resolution of dominant race (in %) from 2020 US Census; (b) tract resolution of median household income (in \$) from 2024 American Consumer Survey; Study areas highlighted in yellow; by Justice Map (Energy Justice Network and Sunlight Foundation, 2023)

To understand the contemporary significance of Puerto Ricans in Holyoke, it is necessary to recognize that Puerto Ricans in the United States have developed a distinctive bicultural orientation. As Maria Pérez y González notes in her book, *Puerto Ricans in the United States* (2000), Puerto Ricans are a bilingual and a bicultural people who have "adapted to their context without necessarily assimilating" (p. 58). Unlike assimilation, which abandons heritage for the mainstream, adaptation selectively incorporates new contexts while preserving core cultural practices. The result is a population that consistently identifies as both Puerto Rican and American—58 percent of Puerto Rican adolescents in one 1994 study perceived themselves as "both" rather than exclusively one or the other (Pérez y González, 2000, pp. 57-58).

This bicultural framework provides a lens for understanding Holyoke's community gardens within the broader arc of Puerto Rican migration, the historical construction of food dependency, and the transnational agricultural movements that have emerged in response. Holyoke's substantial Puerto Rican concentration is the product of multiple waves of migration shaped by economic restructuring, colonial policy, and, most recently, climate disaster. The first significant waves of Puerto Rican migration to Holyoke occurred in the 1950s and 1960s, a period of profound economic transformation for both the island and the mainland. As Margot Wise, a former Farm Manager at Nuestras Raíces, explains, "the mills were already starting to shut down, but there were waves of Puerto Ricans migrating to Holyoke for the cheap housing to work in the surrounding tobacco fields. Puerto Rico is an agricultural country and a lot of people there specialized in that at that time and so they were coming for those jobs" (as cited in McLaughlin, 2018). The migrants who settled in Holyoke were agricultural workers carrying generational knowledge of farming, soil, and crops. When the mills subsequently closed and unemployment rose to nearly 30 percent, this knowledge was preserved in backyards, in shared meals, and eventually, in the community gardens that would emerge decades later.

This pattern of migration and agricultural skill retention is not unique to Holyoke. As documented in the broader scholarship on Puerto Rican settlement in the United States, migrants formed *colonias*, densely settled urban neighborhoods, where they established bodegas, restaurants, and mutual aid societies that served as "support and resource network[s] that aided the transition of those who were established and of the newcomers, and also defended their civil rights" (Pérez y González, 2000, p. 41). This agricultural inheritance is also deeply tied to the forces that systematically dismantled Puerto Rico's agricultural economy and created a condition of profound food dependency. Prior to the 1950s Puerto Rico was an agrarian society with a diverse agricultural base. This began to change dramatically with Operation Bootstrap, an economic development program initiated in 1947 that sought to transform the island's economy from agriculture to manufacturing through tax incentives and industrial promotion (Graf, 2019). The program succeeded in attracting manufacturing investment, but at a steep cultural and nutritional cost. As one account describes it, this "coerced metamorphosis from an agrarian system to an industrialized one eroded traditional cultural practices and even what it meant to be Puerto Rican" (as cited in Graf, 2019). The shift in diet was especially stark, where fresh

vegetables, tropical fruits, and native herbs once grown year-round in the island's tropical climate were increasingly replaced by imported, processed foods. One activist recalls the cultural messaging that accompanied this transition: "We were taught that growing food in your backyard was uncivilized... and we were told that having canned food made us first-class citizens" (as cited in Graf, 2019). The economic consequences were similarly severe as Puerto Rican farm sales decreased by almost two-thirds between 1959 and 1964, according to USDA data (Graf, 2019; National Agricultural Statistics Service, 1962).

By the time Hurricane Maria made landfall in 2017, the island was importing 85 percent of its food (Graf, 2019). This is a dependency that proved to be a crisis when the storm destroyed an estimated 80 percent of the island's crops and farmland, causing the island to import 95 percent of its food in the hurricane's aftermath (Graf, 2019; Rodríguez-Cruz et al., 2022). The nature of food insecurity in the wake of Hurricane Maria differed fundamentally from the chronic food insecurity typically measured by federal agencies. A pilot survey study conducted in Puerto Rico after the storm found that post-disaster food insecurity is characterized by "sudden onset" and "temporary nature," driven not primarily by income loss but by supply chain disruptions and infrastructure destruction such as grocery store closures, lack of food in stores, and physical barriers to access (Mark et al., 2024). Among the twenty-two respondents in the study, the most commonly cited causes of food insecurity were closure of a local grocery store (nine respondents) and "no certain food supply in store" (seven respondents) (Mark et al., 2024, p. 3). Significantly, the study also revealed that the duration of food shortages was a critical factor in how individuals experienced and categorized their own food insecurity. Even respondents who did not self-identify as food insecure reported shortages of specific food groups like protein, dairy, fruits, vegetables but for shorter periods than those who identified as food insecure. This finding suggests that in post-disaster contexts, food insecurity is not a binary condition but a spectrum shaped by time, and that interventions must be responsive to the temporal dimensions of scarcity.

In the aftermath of Hurricane Maria, the island's catastrophic food dependency gave rise to a concerted movement for agricultural self-determination that I believe also extends to the Puerto Rican communities in the mainland US. This farming revolution, as it has been called, was not a spontaneous reaction but the acceleration of a nascent movement that had been

building for years (Graf, 2019). Organizations like El Departamento de la Comida, founded by activist Tara Rodriguez Besosa in 2010, had already begun working to reintroduce native produce to Puerto Ricans through community-supported agriculture models and distribution networks for small farmers (Graf, 2019). The goal, as Rodriguez Besosa articulated, was not merely to increase local food production but to use food as a vehicle for cultural reclamation. She states, "we want to eat the original Puerto Rican diet... we also want to use food to create a better widespread understanding of what truly makes us all Puerto Rican" (as cited in Graf, 2019).

The farming revolution is fundamentally a practice of cultural and historical preservation. In the community of San Salvador, Puerto Rico, as local farmer Carmen Veguilla explains, "farming is a practice of cultural and historical preservation" (as cited in Graf, 2019). The garden in San Salvador, rejuvenated after the hurricane, serves as both a productive landscape and a living archive of Taino heritage, Spanish colonial influence, and African diaspora traditions.

What is often overlooked in accounts of Puerto Rico's farming revolution is that it has a parallel and intimately connected expression in the diaspora. Nowhere is this more evident than in Holyoke, where Nuestras Raíces and other local organizations have spent decades building the very infrastructure of food sovereignty that was resurfaced to attention after Maria's destruction in the island. As Margot Wise recounts, "we've had 3,000 plus people migrate to Holyoke since Hurricane Maria and let me tell you those people aren't coming here to eat kale. They're coming to Holyoke because they have family in Holyoke and it feels familiar and they can sleep on somebody's couch or get by for a while" (as cited in McLaughlin, 2018). Aji dulce, a sweet pepper that is the essential base of sofrito, is not readily available in conventional grocery stores. It is a crop that requires specific knowledge to cultivate and that carries deep cultural meaning. As Wise notes, "people in Holyoke do want to buy vegetables. If you ever think differently, that's [referring to the idea that Holyoke residents choose not to buy fresh produce] a myth; I know that from working at the market" (as cited in McLaughlin, 2018). The post-Maria migrants' need for aji dulce and other familiar foods is a way of maintaining identity and connection in the face of displacement.

This historical and transnational context has direct implications for the present study. The community gardens of Holyoke are better understood as nodes in a *transnational* network of cultural preservation, political resistance, and agricultural knowledge transmission. They are places where the trauma of Hurricane Maria meets the resilience of a community that has been building alternative food systems for decades.

B. Site Histories

Nuestras Raíces oversees nearly twenty gardens across Holyoke. The seven selected for this study listed in Table 1 and mapped in Figure 2 represent the range of governance models, planning contexts, and temporal depths present in the organization's network.

Table 1: Selected Nuestras Raíces Gardens

Garden	Location	Year Established	Governance Model
La Finquita	South Holyoke	1992	Rent-a-plot; organizational management
Cuenta Conmigo	South Holyoke	2001	Rent-a-plot & communal infrastructure
La Piedra	Churchill	c. 2000s	Rent-a-plot; purchased land (2023)
Falcetti Towers	Churchill	c. 2000s	Resident association; HHA partnership
Youth Garden	South Holyoke	2011	Youth education; organizational management
Hidden Treasure	Beaudoin Village (West Holyoke)	2005	Resident plots & school club; HHA partnership

El Renacer	Jarvis Heights (West Holyoke)	2009	Resident coordinator model; HHA partnership
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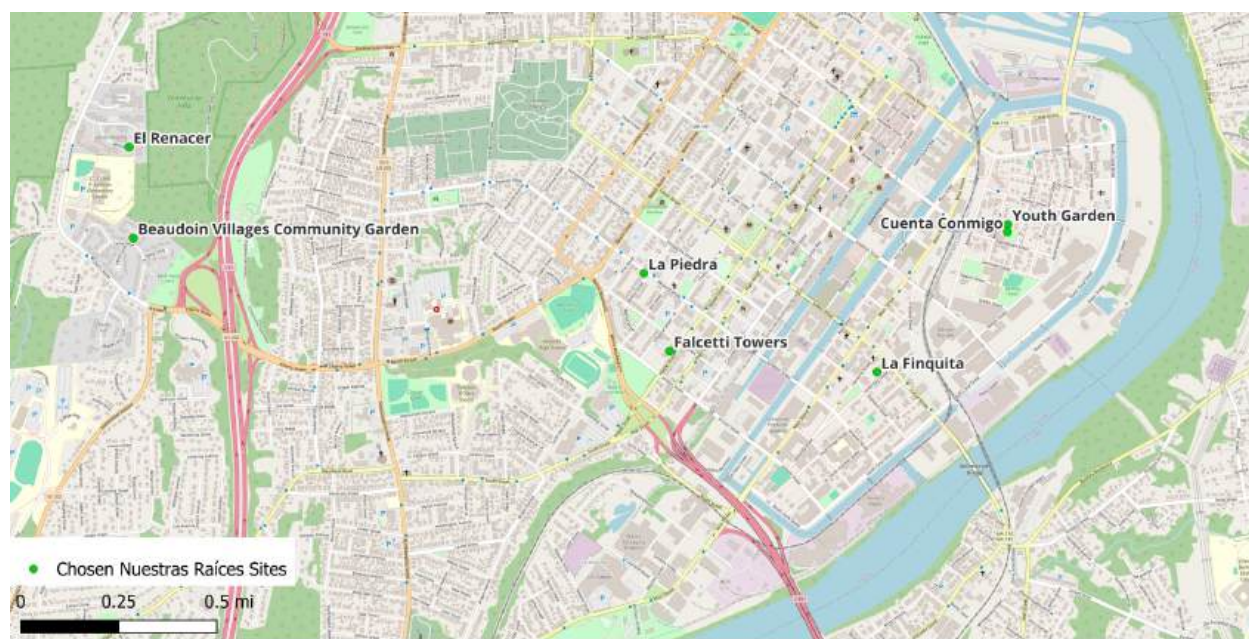


Figure 2: Chosen Nuestras Raíces Sites

The organization's oldest garden, La Finquita (1992), located at Cabot and S. East Street, was established on a former parking lot owned by Providence Ministries. With 32 family plots yielding thousands of pounds of produce annually, it serves as both a productive landscape and a cultural anchor, featuring casitas and staple crops like aji dulce and gandules that reconnect growers to Puerto Rican culinary traditions. Expanding from this foundational site, Nuestras Raíces developed Cuenta Conmigo (2001) in partnership with the City of Holyoke at 99 N East Street. This garden formalized community gathering through its permanent lechonera, stage, and annual festivals, effectively transforming underutilized urban space into a vibrant site of celebration and social cohesion. Its success encouraged the city to develop an adjacent children's park, demonstrating how gardens can catalyze broader public investment in green infrastructure. Nearby, the Jardineros del Futuro Youth Garden (2011) at 116 West Street extended this collaborative model to focus on youth education and skill-building, though it faced periods of inactivity post-pandemic before being revitalized in 2024.

Housing authority partnerships have been central to gardens such as the Hidden Treasure Garden at Beaudoin Village (2005) on Little Avenue and El Renacer at Jarvis Heights (2009) on Gerard Way. These sites illustrate both the potential and fragility of institutional collaborations. Hidden Treasure, shared with Sullivan School, thrived for years as a dual-purpose space for resident plots and youth programming until a 2019 fire disrupted operations. Recent restoration efforts have reintroduced indigenous and Taino practices, signaling adaptive renewal. In contrast, El Renacer's viability has fluctuated with the engagement level of resident coordinators, and the garden is currently fallow despite ongoing efforts to reactivate it –highlighting how relational continuity is as critical as physical design.

Land tenure security also varies across the network. La Piedra, at Sargeant and Walnut Street, began as a leased lot from Hammerman Hardware before being purchased outright by Nuestras Raíces in 2023 through settlement funds, ensuring long-term community control. Meanwhile, Falcetti Towers Garden on Maple Street represents a senior-focused initiative led by the resident association, providing elderly gardeners with access to growing space near key amenities.

CHAPTER FOUR: PLANNING & SPATIAL ANALYSIS

A foundational, community co-design action plan is Holyoke's 2012 Urban Renewal Plan (URP), "Connect. Construct. Create.," which delineates redevelopment priorities across the city's downtown area –where most Nuestras Raíces gardens are located.

Holyoke is a densely populated city with most of its residents in the urban core, industrial, commercial, and suburban half of the City between the river and mountain range. Within this half are the Nuestras Raíces community gardens. This Urban Renewal Plan focuses on the urban core situated at the bend of the Connecticut River, containing South Holyoke. The suburbs are mostly located in West Holyoke, adjacent to Ingleside, with commercial parks like Holyoke Mall. While the urban core of Holyoke contains most of Nuestras Raíces' gardens, the West Holyoke area, *not* included in the Urban Renewal Plan, contains the residential gardens of Beaudoin Village and Jarvis Heights. This distinction arises from characteristics of the urban core or "Center City" that lend to clearance and rehabilitation needs. For example, South Holyoke notably contains many contiguous parcels that are functionally obsolete, and buildings deemed a safety concern. As a result, URP actions include investments in public/open spaces; improving housing options (quality and variety); improving quality of life via the open space network and cultural/creative industries; and increasing property values.

The Holyoke Redevelopment Authority (HRA) identified 10 project areas seen in Figure 3, of which 2 contain Nuestras Raíces gardens: La Finquita in Area 8, near the intersection of Main St. and Cabal St; Falcetti Towers in Area 3, between Maple St. and High St.; La Piedra in Area 3 on Sargeant St. Cuenta Conmigo and Youth Garden are within the Urban Renewal Plan Area, yet not included in the proposed URP sub-areas.¹

¹ I will also note that Lyman Terrace, a garden under Nuestras Raíces that I have not included in this study, is in Area 4, and Dwight Street Garden, a substantial garden not within Nuestras Raíces, is in Area 1.

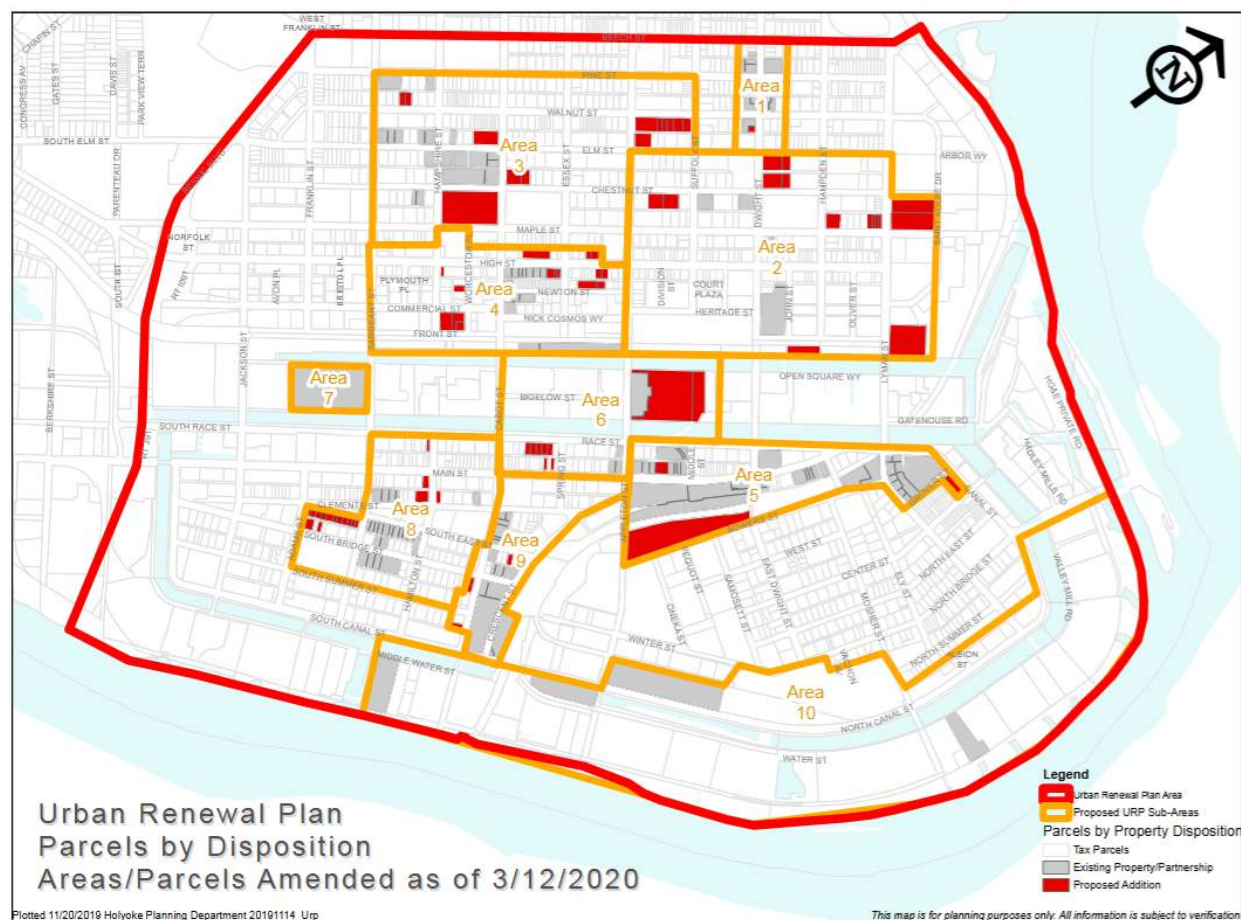


Figure 3: URP Area Map, updated in 2020 (City of Holyoke, 2020)

HRA ranked the project areas within three time frames (short-term or 1-5 years, mid-term or 1-10 years, and long-term or 1-20 years) corresponding to the time it will take to have the most activity realized. The short- and mid-term projects are intended to be the proof of concept for the long-term projects – meaning, the high-risk projects are likely going to be spread out over more years (City of Holyoke, 2012a). Of the areas that coincide with Nuestras Raíces gardens, Area 3 is under Long-Term Expected Completion, and Area 8 is under Mid-term Expected Completion.

The URP public input process conducted at the neighborhood level, by way of establishing a Citizen Participation Committee (CPC), reveals citizen opinions that are described as consistent with the proposed URP actions (City of Holyoke, 2012b). Area 3 falls under the

Churchill Neighborhood and Area 8 falls under the South Holyoke Neighborhood. The actions proposed in Area 3 consist mainly of infill housing, or the process of constructing new, ideally mixed-use and mixed-income², houses on vacant parcels near central social institutions: the Holyoke Public Library, a school, and a senior center. The CPC expressed concern about the consistency between this new variety of housing and existing, adjacent housing. The URP claimed that this would be addressed by engaging in partnerships when planning the housing options. According to the 2020 Major Plan Update (City of Holyoke, 2020a), a parcel (about a 13-minute walk from La Finquita) is to be redeveloped for housing because the "current auto-related business is not compatible with the residential character and planned development of the area." The aesthetic character of the neighborhood as a whole seems dependent on the preservation of numerous historic landmarks in Area 3, including the Holyoke Public Library, Masonic Temple, and a few churches. The update also explicitly mentions the goal of improving the educational experience of the neighborhood.

The actions proposed in Area 8 build off of the 2008 South Holyoke Revitalization Strategy, whose goal was to create new mixed-income housing. The URP added goals for new open space near the Morgan School and neighborhood retail. The CPC affirmed the priority of mixed-use housing and home ownership opportunities and emphasized efforts to improve the connectivity of the neighborhood using walkways. There were also concerns about crime and safety. The contracted work from the Holyoke Housing Authority, as a result, led to the concept of South Holyoke Homes, which continues the neighborhood's desire for more housing around Carlos Vega Park (a 13-minute walk from Falcetti Towers and a 5-minute walk from La Piedra). The 2020 update for Area 8 identified many parcels – mostly parking lots – of blight and neglect. Similar to Area 3's updates, a parcel for "auto use" was regarded as not compatible with the area's residential character, and the update specifically cites the goal of increasing local property values. Notably, the update cites "visibility" as the reason for targeting corner parcels.³

² Mixed-income refers to combining market-rate and affordable units (often 20%–60% of units) in single developments.

³ As an aside, the concerns in Area 3 and 8 seem vastly different from those of Area 1, consisting of the Dwight Street Garden. Area 1 has more actions related to streetscape improvement, responding to CPC concerns for blighted appearance and a request for more art and visual elements.

This neighborhoods' emphasis on mixed-use, infill housing and enhanced walkability within the urban core directly intersects with perhaps the most planned feature of a community garden –neighborhood proximity. The very logic of community gardening assumes service to a proximate residential population. A spatial analysis of garden locations (Finnell et al., 2022) in relation to land use data (later in Figure 5) reveals that they are strategically concentrated in neighborhoods with high residential density and, notably, higher rates of home rentership in multi-family units.

Applying a 0.25-mile walking distance (a standard pedestrian-shed metric) around each garden location maps the residents they are formally designed to serve most easily. The mention of more "housing choices" refers to including multi-family, townhouse, and duplex styles that complement existing neighborhood character. While the plan prioritizes mixed-housing infill, the practical outcomes –whether through new construction, rehabilitation, or rising property values –could inadvertently alter the demographic or economic profile of these neighborhoods over time. In areas that do opt for more single- or double-family housing as opposed to multi-family, the underlying spatial logics of the URP and the gardens may diverge: one aiming to increase property values and homeownership in a reconfigured streetscape, the other aiming to provide accessible, use-oriented open space for existing, often renter-dominated communities. Thus, the "importance" of garden proximity, as planned, hinges not just on current residential patterns but also on how successfully the URP's vision for housing variety preserves access and stability for the existing renter communities.

The URP's focus on diversifying housing types within the urban core is part of a longer planning arc that also explicitly valued community gardens as a means to enhance neighborhood quality of life. This dual focus on housing and gardens suggests a planned intent to support dense, populated neighborhoods through both built form and social infrastructure. Holyoke's 2009 Center City Vision Plan (CCVP) laid this foundational framework, establishing six principles (Massachusetts Housing Partnership, 2009). Chief among these were increasing density and development types; expanding housing choices with styles that complement each neighborhood; and creating a walkable, safe, pedestrian-friendly city. Crucially, the CCVP's chapter on "Enhance the Urban Neighborhoods" recommended that the City "explore

opportunities to convert appropriately placed and accessible vacant lots into community gardens" to offer residents access to locally grown foods (Massachusetts Housing Partnership, 2009, p. 32). This marks a formal, early recognition within the planning continuum of community gardens as a legitimate neighborhood asset and a component of open space and food access strategies.

The Beaudoin Village and Jarvis Heights gardens in West Holyoke operate in a different context. Unlike downtown, West Holyoke was not targeted by the 2009 CCVP or the subsequent 2012 URP. This area, characterized by more suburban residential patterns, did not undergo the same level of public, city-wide visioning or redevelopment planning during this period. Consequently, the planned garden features in these sites cannot be reconstructed from a comparable suite of municipal master plans or urban renewal documents. Instead, their planning logic must be inferred from a different stratum of institutional documentation: the project-specific plans and portfolios of the Holyoke Housing Authority (HHA) and related affordable housing developers. While comprehensive area-wide development plans for West Holyoke were not identified for this study, the HHA's broader portfolio during this era—including major developments such as Beaudry-Boucher Apartments, Churchill Homes, and various elderly, high-rise, and scattered-site housing—reveals a consistent institutional mission. This mission prioritizes providing quality affordable housing; fostering resident stability and community; and enhancing property management through amenity spaces that improve quality of life.

Thus, for Beaudoin Village and Jarvis Heights (and also Falcetti Towers, which is an HHA development), the "important" garden features are likely framed not within a narrative of city-wide economic revitalization or blight removal, but within a housing development and management paradigm—particular that of federally assisted housing. This distinction in planning lineage between city-led redevelopment in the core and housing authority-led development in the suburbs establishes my comparative axis for the study. It posits that the very definition of what makes a garden feature "important" differs fundamentally based on the institutional actor (HRA vs. HHA) and the spatial logic (revitalizing a distressed core vs. managing suburban-style affordable housing) that governs its planning. This framework sets the stage for assessing

whether gardens designed as housing amenities manifest different forms of latent capacity or vulnerability during a crisis than those conceived as tools for urban revitalization.

While housing authorities in the United States have traditionally functioned almost exclusively as rental developers and managers, the HHA charted a distinct path in the urban core by strategically integrating affordable homeownership into its South Holyoke redevelopment project. This initiative responded directly to a documented neighborhood condition where over 90% of homes were renter-occupied and to a longstanding community advocacy for ownership opportunities (Forman et al., 2025). Consequently, rather than encountering the resistance that often greets new housing proposals, the HHA's plan garnered strong local support, as residents welcomed the conversion of long-vacant, blighted lots into assets that would build stability and intergenerational equity for local families. To achieve financial feasibility, the HHA utilized modular construction, a method that lowered costs and accelerated timelines while still delivering duplex-style homes that balanced necessary density with the community's preference for a single-family-like neighborhood character. These residential gardens represent a notable case of a housing authority acting as a catalytic developer to expand homeownership in a manner that was both financially viable and embraced by the existing community.

The West Holyoke gardens were established in the early 2000s, a period preceding the HHA's entry into the federal Moving-to-Work (MTW) demonstration program in 2013. Therefore, while the current 2026 MTW Annual Plan reveals the contemporary managerial logic governing these sites (Holyoke Housing Authority, 2025), the gardens' original planning must be situated within the HHA's earlier development priorities and the federal housing policy landscape of the early 21st century. The MTW framework, solidified after 2013, provides the clearest lens into the ongoing institutional priorities that now shape garden maintenance and capital planning. The program's core statutory objectives –to reduce costs, increase housing choice, and promote economic self-sufficiency– create an operational environment where physical amenities are secondary to housing management goals. This is evidenced by the plan's sole direct reference to a garden site, Beaudoin Village, which appears as a line item for capital expenditure ("Lead Compliance," "Site Improvements") rather than as a community program (Holyoke Housing Authority, 2025, p. 11). Furthermore, the plan's extensive focus on

administrative streamlining (e.g. Biennial Recertifications) and targeted voucher programs (e.g. for youth aging out of foster care) illustrates a system optimized for transactional efficiency and demographic targeting, not for nurturing the kind of open-ended, community-managed social space that a garden represents (Holyoke Housing Authority, 2025, pp. 6-8, 13).

Consequently, the important features of these gardens, as perceived through the contemporary MTW lens, are likely those that align with this derivative, managerial logic. This includes compliance and risk mitigation, related to design features that consider zoning regulations, regulations on the public sale of produce, land use policies related to water access and plot sizes, and benefits to entering in partnerships with local non-profits (Intact Public Entities, 2019); low operational burden, using layouts and plantings that require minimal intervention from HHA staff and aligning with a cost-effectiveness mandate; and amenity value, emphasizing aesthetic contributions to property appeal, supporting tenant retention and housing choice.

A garden conceived in the early 2000s as part of a community development or placemaking initiative (potentially aligned with then-prevalent HOPE VI or community revitalization ideals) is now managed within a system structured for asset management and administrative efficiency. This disconnect posits that the latent capacity of these gardens may be rooted in an original social intent that is no longer the primary focus of the stewarding institution, potentially rendering them more susceptible to neglect if their social functions are not actively maintained by residents themselves.

In the case of Falcetti Tower, the planning context is uniquely defined by its status as a Designated Elderly-Only public housing high-rise, as formally established in the HHA's 2012 Elderly-Only Designated Housing Plan (DHP). This document transcends general urban renewal goals, providing a precise institutional rationale for the building and, by extension, any amenities within its grounds.

The DHP's justification is linked to City of Holyoke Consolidated Plan priorities to "increase availability of elderly units" and to "address community concerns such as supportive services that increase the ability of seniors... to live independently" (Holyoke Housing Authority,

2012, p. 2). The plan's objectives for designating Falcetti Towers as elderly-only suggest an emphasis on, firstly, fostering community and reducing isolation. This positions the garden not merely as a green space but as a purposive social infrastructure designed to mitigate the risks of social isolation. Another goal is to enable independent living. A garden can be framed as a therapeutic landscape that promotes physical activity, mental well-being, and engagement with a manageable outdoor environment, supporting this overarching goal. Further, it aims to facilitate service delivery and safety. This implies that the garden's design must support legibility, accessibility, and oversight, aligning with principles of Crime Prevention Through Environmental Design (CPTED) tailored for an elderly population.

This DHP reveals a nuance that the garden at Falcetti Towers may have success metrics framed in terms of resident social engagement, perceived safety, and quality of life. This distinct institutional purpose must be accounted for when evaluating its "performance" and latent capacity, as its design logic prioritizes social cohesion and accessibility over productive yield or political mobilization.

While the municipal CCVP and URP establish a broad redevelopment framework, and the HHA documents define property-specific management logics, a critical third layer of planning explicitly links these citywide goals to the grassroots food systems work of Nuestras Raíces. This linkage is formalized in the 2017 Community Action Plan (CAP) for Holyoke, Massachusetts, developed through the federal Local Foods, Local Places (LFLP) technical assistance program, a collaboration between the EPA, USDA, the City of Holyoke, and Nuestras Raíces.

The plan originates from a 2016 request by Nuestras Raíces for assistance "to develop an action plan to grow the city's food economy, to increase access to healthy food, and to include food production, food businesses, and food culture in the planned redevelopment of the city" (EPA, 2017, p.3). The CAP translates municipal priorities into actionable food sovereignty goals. Most salient for this study is Goal 3: "increase production of food within the city of Holyoke", which cites community gardens and includes actions to replicate supportive ordinances and better utilize land for food production (EPA, 2017). This goal seeks to partner with local

institutions and mainstream the term small-scale 'growers' rather than farmers or producers, signaling a deliberate effort to validate culturally-specific, non-commercial agricultural identities. This production focus is complemented by Goal 4, which aims to "increase opportunities for farmers, growers, and food businesses to market their products" and explicitly identifies South Holyoke as a USDA-defined food desert, prioritizing access to fresh, culturally appropriate produce (EPA, 2017). Together, these goals frame Nuestras Raíces' gardens, particularly those in South Holyoke like La Finquita, not merely as open spaces, but as critical infrastructure for food production, cultural identity, and economic mobility within a disinvested neighborhood.

Furthermore, the plan acknowledges the practical reality shaping these gardens' development, where all but one of Nuestras Raíces' nearly 20 properties are owned by the city. While these sites – a form of non-profit recreational parcels – are not likely to be taken out of recreational use, the parcels for community gardens are vulnerable to development and displacement, as seen with Lyman Terrace in 2017 (The Conway School, 2018). Lyman Terrace was a former partnership with the HHA and ended up being redeveloped into more housing after the property failed to be up to code due to funding issues. This condition of land tenure directly ties the organization's fate to municipal planning decisions, making the alignment demonstrated in the URP and this CAP not just strategic but essential for the gardens' survival and legitimacy. Thus, the CAP serves as a vital translational document. It absorbs the municipal redevelopment logic of the URP and reframes it through the lens of food justice and community self-determination. This provides the specific criteria against which the performance of Nuestras Raíces' gardens, as agents of food sovereignty within a city-led renewal area, can be most meaningfully assessed. Table 2 consolidates these criteria, aligns them with feasible garden features, and maps them to the relevant case study sites.

Table 2: Holyoke's Municipal Planning Criteria and Corresponding "Important" Garden Features

(Municipal) Criteria	Important Features	Site(s)
Improving facilities for	Spaces for educational collaboration	Falcetti Towers, La Piedra

educational opportunities while transitioning surplus educational buildings. (URP Area 8; CCVP "Learning Node")	(e.g., student-friendly gardening beds, interpretive signage, designated workshop/class areas).	
Enhancing connectivity & walkability (URP;CCVP)	Proximate, visible location within a standard pedestrian shed (e.g. 0.25-mile radius); clear, safe pedestrian access from sidewalks.	All gardens in the urban core (La Finquita, Falcetti Towers, La Piedra, Cuenta Connigo, Youth Garden)
Increasing housing choice & neighborhood density with context-sensitive infill. (URP;CCVP)	Siting that serves existing high-density, renter-occupied residential areas. Integration with housing complexes	La Finquita, Falcetti Towers, La Piedra
Converting vacant/blighted lots into community assets. (URP; CCVP "Enhance Urban Neighborhoods" directive)	Spaces that reclaim underutilized land, creates "eyes on the street," and demonstrates care through clear boundaries, fencing, and cultivation.	All gardens in the urban core (La Finquita, Falcetti Towers, La Piedra, Cuenta Connigo, Youth Garden)
Fostering neighborhood identity & cultural heritage aesthetically; consistent with residential character. (URP)	Spaces for gathering/ritual; expressive landscapes (altars, art, structures reflecting community values).	La Finquita, Falcetti Towers, La Piedra
Improving quality of life via open space network. (URP action item; CCVP "Expand Open Space Amenities")	Multi-functional Open Space: Provision of both productive plots and social/communal areas (lawns, seating, shaded areas) that function as de facto neighborhood parks.	All gardens in the urban core (La Finquita, Falcetti Towers, La Piedra, Cuenta Connigo, Youth Garden)
Promoting local food systems & food access (CCVP)	Allocation of individual/family plots; presence of communal growing areas for food distribution.	All gardens in the urban core (La Finquita, Falcetti Towers, La Piedra, Cuenta Connigo, Youth Garden)
Addressing crime and improving public safety (URP Area 8)	Open layout, low perimeter planting; controlled access points (defined gates/fencing); natural surveillance through proximity to housing & active communal areas;	Falcetti Towers, La Piedra

	aesthetics of care	
Compliance & risk mitigation, satisfying zoning regulations and land use agreements (HHA)	Limitations on land alterations, watering times, excessive runoff into neighboring properties; reduced turf areas typically used as multifunctional spaces for recreation; self-serve, members-only models reducing casual, unrestricted access	Hidden Treasures, Renacer, Falcetti Towers
Low operational burden, requiring minimal intervention from HHA staff and aligning with a cost-effectiveness mandate (HHA)	Use of native plants, water-wise landscaping, and stormwater management techniques like rain gardens; using governance strategies in partnerships with local non-profits	Hidden Treasures, Renacer, Falcetti Towers
Amenity value for property appeal, supporting tenant retention and housing choice (HHA)	Well-maintained and uniform structures like raised beds; clear pathways and defined edging (gravel, pavers or woodchips/mulch) for visual organization; vertical elements (garden trellises, arbors or pergolas) to add height and a focal point, while also acting as a social, shaded area; welcoming entrance; "pollinator corridors" using flowering plants	Hidden Treasures, Renacer, Falcetti Towers
Fostering community and reducing isolation (HHA DHP)	Accessible & safe social spaces with raised, barrier-free planting beds, seating in shaded areas; clear, well-lit pathways, and open sightlines for passive surveillance; Layouts that facilitate group activities, gardening workshops, or intergenerational events	Falcetti Towers
Supporting independent living for elderly residents in designated housing (HHA DHP)	Low-intensity horticulture, focusing on easy-to-maintain flowers, herbs, or sensory plants rather than intensive food production.	Falcetti Towers

This section of the analysis now shifts to the spatial and material dimension of these sites. I examine how the areas surrounding the gardens, such as the residential or commercial spaces, shape the contexts and potential uses implied by the aforementioned plans.



Figure 4: Housing type and location of garden for (a) Falcetti Towers, (b) Beaudoin Village and (c) Jarvis Heights using Google Maps images

I will begin with the gardens in residential areas of Falcetti Towers, Beaudoin Village, and Jarvis Heights (Figure 4). The housing typology, the form of the residential units, creates distinct relationships between residents, their private domain, and the shared garden space. Falcetti Towers is a high-density apartment tower. This vertical, centralized form creates a large, concentrated population with a single, shared entry/exit. The garden here is an exterior common area for the entire building, akin to a courtyard or yard for a large apartment complex. Its use is inherently collective and public for the building's residents, but its relationship to any individual household is indirect. Beaudoin Village consists of low-rise townhouse-style units. This typology, with individual unit entrances at ground level, creates a more direct, quasi-private relationship between the home and the outdoor space. Gardens in this setting can function more like front or back yards for a cluster of homes, encouraging casual, daily interaction and a sense of territorial responsibility from adjacent residents. However, due to a partnership with the neighboring Sullivan Middle School, the garden was located with closer proximity to the school. Jarvis Heights, while also an apartment complex, presents a mid-rise or garden apartment form different from the centralized tower of Falcetti. This can create semi-private outdoor zones shared by smaller clusters of units, a spatial condition that falls between the collective courtyard of a tower and the direct access of a townhouse. This distinction in domestic spatial logic is critical for evaluating garden "performance." A garden's success in fostering social interaction or stewardship may be heavily influenced by whether residents encounter it as a destination (as from an apartment tower) or as an extension of their domestic sphere (as from a townhouse). The notable design features for each garden –such as seating location, pathway design, plot allocation –would ideally respond to these underlying typologies to effectively serve their resident populations.

This spatial analysis also extends beyond the immediate boundaries of the garden and its associated housing to examine the broader neighborhood context in which each site is embedded. This layer of analysis, drawn from aerial imagery and GIS mapping, reveals the underlying land use patterns and proximity to social infrastructure –key institutions like schools, churches, and parks –that shape a garden's potential role and accessibility within the community.

Figure 5 establishes the primary walkability catchment for each garden, applying a standard 0.25-mile pedestrian shed mentioned prior. This analysis visually demonstrates which residential populations each garden is most physically accessible to, highlighting disparities in service areas between, for example, the densely packed urban core sites and the more dispersed West Holyoke locations.

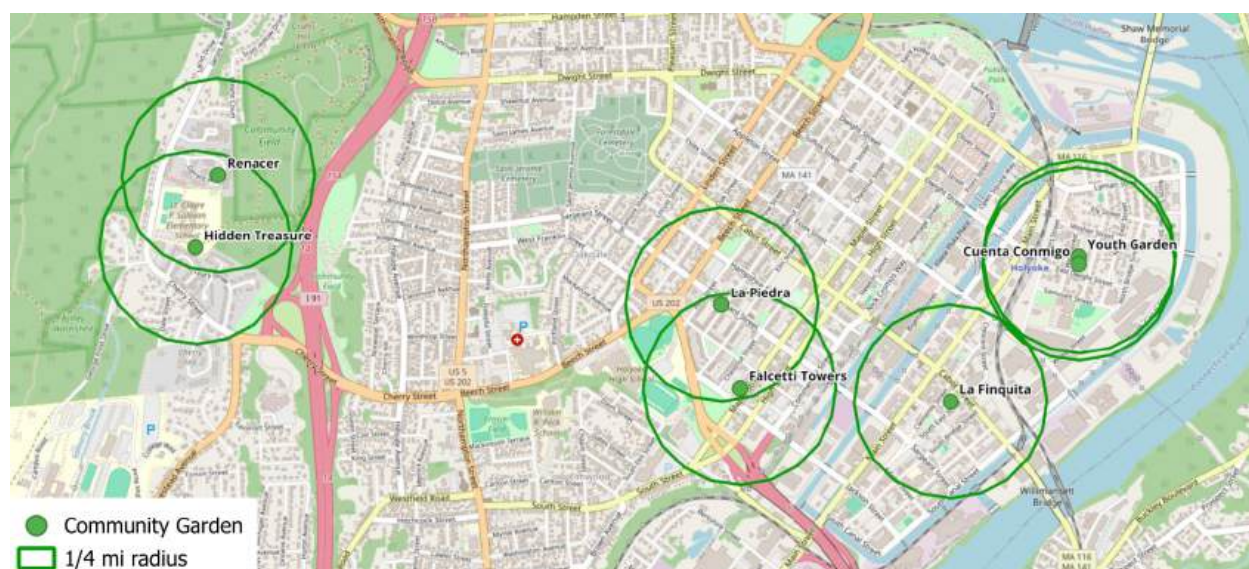


Figure 5: Holyoke study area with site locations & 0.25mi "Ped Shed"; By author

We see that for the urban core gardens (La Finquita, Cuentra Conmigo, Youth Garden, Falcetti Towers, and La Piedra) their catchments overlap considerably, creating a networked field of accessible open space across South Holyoke and Churchill. La Finquita's catchment extends eastward along Cabot Street and north toward Main Street, capturing the dense multi-family housing stock that characterizes South Holyoke's residential fabric. Cuentra Conmigo and the Youth Garden, separated by less than 200 feet, generate a contiguous service area that encompasses Pina Park and reaches toward the Morgan School, effectively doubling the accessible open space within their shared catchment. La Piedra's catchment, positioned along Sargeant Street, bridges the gap between the Churchill and South Holyoke neighborhoods, suggesting its potential as a connective node in the urban core's open space network. Falcetti

Towers, situated at the intersection of Maple and High Streets, anchors the northern edge of this network, its catchment extending toward the Holyoke Public Library and the senior center identified as key civic institutions in the URP's Area 3 planning.

The West Holyoke gardens present a dramatically different picture. Hidden Treasure's catchment at Beaudoin Village is constrained by the surrounding suburban land use pattern: the garden serves a concentrated pocket of townhouse-style units but lacks the overlapping catchments that characterize the urban core. Its service area does not extend significantly beyond the Beaudoin Village development itself, partly due to the interruption of Interstate 91 and the commercial corridors that physically separate this pocket from adjacent residential areas. El Renacer's situation at Jarvis Heights is even more isolated. While the garden is centrally located within the Jarvis Heights development, the surrounding land use—a mix of parking lots, the Jarvis Heights apartment buildings themselves, and the wooded riparian corridor along the Connecticut River—creates a catchment that serves primarily the immediate residents of this single development. Unlike the urban core gardens, whose catchments bleed into one another and into surrounding commercial and institutional spaces, the West Holyoke gardens function as discrete islands of accessible open space, their reach limited by the very suburban morphology.

Figure 6 provides the material and ecological context of each garden. This land cover data reveals whether a garden sits within a matrix of impervious surface (typical of the urban core), adjacent to riparian corridors (like the Connecticut River or canals), or amidst suburban vegetated lots. This environmental embedding influences microclimate, stormwater management potential, and the garden's perceived character as an "oasis" or an integrated part of the built fabric.



Figure 6: Holyoke study area with Mass Land Cover 2016 Data (MassGIS, 2016); By author

The urban core gardens occupy a landscape dominated by impervious surfaces: rooftops, parking lots, and paved streets. La Finquita, established on a former parking lot, sits at the center of a dense grid of residential and commercial buildings, its soil likely contaminated from its prior use, requiring raised beds to become productive. Cuenta Conmigo and the Youth Garden share this condition, though their adjacency to Pina Park provides a rare pocket of permeable surface within the impervious matrix. La Piedra, located at the intersection of Sargeant and Walnut Streets, is embedded in a similarly dense fabric, though its purchase by Nuestras Raíces in 2023 secured permanent control over what was once a leased lot. Falcetti Towers occupies an interstitial position where the garden itself sits on managed lawn adjacent to the apartment tower, but it is bounded on all sides by the dense urban fabric of downtown Holyoke. The surrounding impervious surfaces like the parking lots of the senior center, the paved playground of the adjacent school, the streetscape of Maple Street may contribute to a heat island effect that influences the garden's microclimate. For elderly gardeners, this heat exposure presents a barrier to summer cultivation, a factor that may explain the garden's design emphasis on low-intensity horticulture and shaded seating areas.

As for the West Holyoke gardens, Hidden Treasure, situated at the edge of Beaudoin Village, is embedded in a matrix of suburban vegetation. The surrounding land cover includes significant tree canopy, lawns, and the open space of the adjacent school grounds. This vegetated context moderates temperature, retains moisture, and provides habitat for pollinators –ecological advantages that should, in theory, support productive cultivation. Yet the garden's neglect suggests that ecological suitability alone does not drive performance; the social infrastructure required to activate that ecological potential remains absent. El Renacer at Jarvis Heights occupies the most ecologically complex site of all seven gardens. Located at the edge of the Connecticut River's riparian corridor, the garden is adjacent to wooded areas that provide shade, wind protection, and wildlife habitat. The land cover data reveals a transition zone between the built environment of Jarvis Heights and the natural landscape of the river. This riparian adjacency offers opportunities for stormwater management, pollinator habitat, and ecological education that the garden's current neglect leaves unrealized. Yet this same ecological context also presents constraints where the wooded edges reduce visibility from the surrounding streets, contributing to the garden's hidden character, and the proximity to the river may bring flood risk that institutional planning documents do not address.

The concept of open-space ratio, or the proportion of open space to enclosed space, offers a final lens for understanding these differences. In the urban core, where building coverage is high, each community garden makes a proportionally larger contribution to the neighborhood's open-space ratio. A vacant lot converted to a garden in South Holyoke may double or triple the accessible open space within a quarter-mile radius. In West Holyoke, where open space is more abundant but less accessible due to the dispersal of residential units, the same garden contributes less to the neighborhood's overall open-space ratio but may serve as the only accessible green space for residents without cars. This inversion of dense neighborhoods with too little open space versus suburban neighborhoods with open space that is poorly distributed suggests that the ecological context of each garden must be understood in relation to the social context it serves.

To aid with that social context, Figure 7 (Holyoke sites & neighboring social infrastructure) maps the gardens in relation to schools, churches, and parks. The clustering of these civic assets around the urban core gardens creates conditions for what I term

"network-forming tendencies" or the capacity of a garden to serve as a node within a broader constellation of civic institutions.



Figure 7: Holyoke sites & neighboring social infrastructure (schools, churches, and parks); By author

For instance, a garden located adjacent to a school (La Piedra & Falcetti Towers near Morgan School; Hidden Treasures near Sullivan Middle School) is positioned for formal and informal educational programming. A garden proximate to a church may align with cultural and ritual practices. A garden proximate to a church may align with cultural and ritual practices. The South Holyoke gardens are situated within walking distance of several congregations like St. Peter's, Our Lady of Guadalupe, and others that serve as anchors of Puerto Rican cultural and religious life. This adjacency suggests the potential for gardens to function as extensions of these institutional spaces, hosting community meals, religious festivals, or mutual aid networks grounded in the congregational infrastructure. Gardens near public parks (La Finquita near Carlos Vega Park or Cuentra Conmigo and Youth Garden near Pina Park) may function as complementary productive or social nodes within a larger open-space network, rather than as isolated amenities. This complementarity operates in two directions: the park provides the lawn, playground equipment, and open space that a garden focused on cultivation lacks, while the garden provides productive, culturally specific agriculture that a conventional park does not

offer. Together, they create a field of social infrastructure with greater functional diversity than either space alone could provide.

A garden's design and governance can either leverage or neglect these adjacencies. For example, a garden gate that opens toward a schoolyard invites a different relationship than a garden surrounded by fencing on all sides. A bench positioned to view a park encourages different social behavior than one tucked against a shed. A governance structure that includes representatives from adjacent institutions creates different accountability than one that treats the garden as a bounded entity. Thus, the analysis of neighboring social infrastructure provides a spatial variable to the municipal criteria of "improving educational opportunities" or "enhancing open space networks," (see Table 2) allowing for a more grounded assessment of how planned importance is, or is not, materially facilitated by location. Whether these adjacencies become activated or remain dormant—a question this analysis raises but does not resolve—will be taken up in the next chapter, where the material traces of use, stewardship, and adaptation reveal which gardens have successfully leveraged their civic context and which have not.

Figure 8 extends the scope of social infrastructure to include Holyoke food resources as mentioned in the 2024 Holyoke Community Cookbook (Hanna, 2024), adapted from a resource developed by the FoodBank of Western Massachusetts. This mapping situates the Nuestras Raíces gardens within the city's broader food access landscape, revealing the institutional and grassroots resources that could be activated in crisis response.



Figure 8: Holyoke sites & other food resources; By author

CHAPTER FIVE: SITE OBSERVATIONS

Chapter Four reconstructed the planned "importance" of Holyoke's community gardens – how municipal documents framed some gardens as revitalization tools and others as housing authority amenities. This chapter moves from what was planned to what exists. I document the material traces of seven Nuestras Raíces gardens across Holyoke, classifying each as Lively, Alive, or Neglected based on evidence of stewardship, social infrastructure, and cultural practice. I then extend the analysis to two comparative cases: Gardening the Community, an infrastructurally-focused, youth-led hub-and-spoke model, and Holyoke Food & Equity Collective, a practice-based, landless mutual aid model.

A. Nuestras Raices

The material traces at Renacer (in Jarvis Heights) reveal a garden shaped more by institutional neglect than community action. The site is dominated by perennial plants and weeds with no observable edible annual crops. The most telling trace is the brightly colored shed: its doors are barely hanging on their hinges, secured by a chain and padlock –a physical testament to infrequent use and failed maintenance.



Figure 9: Renacer Shed

The garden gate, also padlocked, shows minimal ground disturbance, indicating limited access rather than fluid community entry. The site lacks user-built structures, seating, or

decorative elements. The original parceling is obscured by overgrowth, and there is no clear pathway to the shed and bench in the far corner. By virtue of its central location in the radial streets within Jarvis Heights, it seems to be a members-only model with hidden access. At its peak performance, the garden seems more like a visual amenity than a productive space. Adding to its hidden location in the residential area, it is not in an area with a lot of recreational pedestrian traffic. Rather, it is central to various parking lot spots and its visibility is challenged by the numerous parallelly-parked cars around it. Even the garden's sign is obstructed by cars, making it function more as a token green space than a social hub amidst an open-space deficit in the residential area.



Figure 10: Renacer surrounding pathway

The neighboring pathway is singular and only serves as a perimeter for the garden. This reveals a trade-off between occupying an underutilized vacant space (what would've otherwise been more parking spots) and being front-facing, surrounded by open space that would invite pedestrian and recreational traffic. By virtue of being in a parking lot, pedestrians are restricted to those with a different destination in mind, hoping to leave the residential area instead of embracing the nature in it. The reclamation of underutilized land really only works in a rent-a-plot model that is not dependent on organic assembly. These traces collectively illustrate a gap between the municipal vision of a compliant, low-burden amenity with "well-maintained

uniform structures" and the reality of a low-burden amenity that invites no communal investment or adaptation.

At Hidden Treasures (in Beaudoin Village), material traces point to a breakdown in the intended partnership between a school and a housing community. The most striking feature is the eroded half-Puerto Rican, half-American flag discarded on the fence—a symbolic artifact of fragmented cultural identity and waning institutional pride.



Figure 11: Hidden Treasure sign found adjacent to main entrance

Hidden Treasures, like Renacer, have signs of minimal cultivation. In terms of access, there is a direct pathway between the back of Beaudoin Village and Clare Sullivan Middle School, on which the garden is at the midpoint. It is understandable that the middle school cannot provide a front-facing entrance to the garden for safety concerns; however, the location of this garden relative to the residents is arguably worse than in Jarvis Heights. At least Jarvis Heights had pedestrian traffic, albeit going towards their cars. In this case, despite there being a clearly defined walkway, there are no neighboring amenities or open spaces that would invite pedestrian traffic beyond traveling to the school. This might decrease the pressure for visual organization and cohesiveness, as evident by the lack of delineation between the school side of the garden and the resident side. Further, the school projects lack an identity: comparing this with Hancock Street Farm in Springfield (in Chapter Five), whose students engage in creating their own custom signage, there is no visual evidence that students were here.



Figure 12: Hidden Treasures Residential Side



Figure 13: Hidden Treasures Student Side

It is multifunctional in the sense that there is no delineation of vegetation beyond the students' garden beds. As a result, the resident side is littered with small, spontaneous gardening attempts in tires and scattered trash, indicating a lack of coordinated stewardship. This is reflected in the focal point of the garden which is the dilapidated shed at the center, shared by both the student and residential sides.



Figure 14: Hidden Treasures shed

These deteriorations highlight a severe gap with municipal goals for "well-maintained structures," "visual organization," and "spaces for educational collaboration." The garden was planned as a shared community asset but performs as an unclaimed liminal zone with no architectural identity or invitation.

La Finquita's material traces emphasize utilitarian organization over communal expression. The garden is laid out in a visible grid of about 30 rented plots, a formal structure imposed by the organization that standardizes individual/family use. Notably, there are no user-built structures –no sheds, shaded seating, decorative fencing, or cultural markers within the garden itself. The perimeter is defined by hedges and vegetation that cover the fences, reducing visibility from the street and creating a somewhat insular feel.



Figure 15: La Finquita front entrance

The primary interaction occurs outside the garden: an adjacent multi-family complex where residents were observed congregating over drinks in a dedicated courtyard space. This separation reveals a gap where, while the garden successfully performs as a productive landscape reclaiming a former parking lot, it lacks the design elements of social infrastructure – like stages, kitchens, or gathering shelters – envisioned in plans to "foster neighborhood identity and cultural heritage aesthetically." The architecture here facilitates individual food security but does not materially support collective life or cultural expression.



Figure 16: La Finquita adjacent residential complex



Figure 17: La Finquita signage

Evidenced by signage in La Finquita, there is an organizational focus on formal knowledge production. These formal ways for producing practical knowledge include free

workshops, lectures, afternoon programs and summer camps. As seen in these signs, we observe a Soil Health Workshop on the left, whose purpose is "to learn about sustainable ways to maintain the health of your garden's soil. This event is free and open to the public." Similarly, on the right is a Community Garlic Planting event to "learn how to plant garlic for next year's harvest."

The material traces at Falcetti Towers reflect a design logic centered on accessibility, safety, and low-intensity use for an elderly population. The garden features raised beds that are fenced with wire barely visible from afar—a design prioritizing barrier-free access and subtle boundaries. There are no user-built structures, shaded pavilions, or communal cooking facilities. The architecture is minimal, functional, and institutionally provided. It meets accessibility standards but does not invite user modification or collective placemaking. The multifunctionality and "de facto" park status of this garden comes from its adjacency to a park or other green spaces, allowing it to benefit from the recreation-seekers that may be near the garden. In fact, the neighboring park infrastructure seeps into the Falcetti garden as they both share the same green benches. The proximity to housing and the broader public is assured by the garden beds' location in and in front of the apartment complex. This contrasts with the other residential gardens (Jarvis Heights and Beaudoin Village) which host their gardens either central to the residential area as in the case of the radial pathways of Jarvis Heights, or hidden as in the case of Beaudoin Village.



Figure 18: Falcetti Towers garden beds

These traces align closely with the narrow municipal goals for this site: "low-intensity horticulture" and "supporting independent living for elderly residents." However, they reveal a gap with broader resilience planning that values adaptable, user-generated spaces capable of fostering mutual aid or cultural exchange. The architecture here fulfills a compliance-focused, amenity-driven mandate but does not cultivate the spatial agency or communal identity seen in more vibrant gardens.

Cuenta Conmigo presents the richest set of material traces, indicating a lively, culturally embedded space where vernacular architecture embodies collective, user-driven creation.

Approximately 15-20 active rental plots host diverse vegetation, including microgreens, green peppers, cover crops, aji dulce, and citrus. The combination of cover crops and produce suggests an active crop rotation system for soil health. The garden is physically defined by permanent social infrastructure: a lechonera (barbecue pit), a stage, and distinctive blue picket fences.



Figure 19: Cuento Conmigo main entrance from Pina Park



Figure 20: Cuentra Connmigo social infrastructure

"We had lunch at the lechonera" is a typical phrase in Puerto Rico; "...every Saturday on roads through the countryside you can smell the sweet smoke of a lechonera on your way to the beach with family" (Chitwood, 2022, p. 925). Lechon is a typical Puerto Rican dish featuring pork, an ingredient with deep ties in Spanish identity and society. A space catered to meals beyond what is produced at the farm is an indication of multifunctionality. Cuentra Connmigo does not have to act as a "de facto" neighborhood park as there is a colorful, eye-catching children's playground directly adjacent to it (Pina Park). There is also a "casita" which is a wood-frame structure or "little house." Casitas, typically with swept bateys or open spaces around them, have architectural roots in the structures of indigenous Taino Indians. These casitas host eating as a group affair. Bateys in particular are sacred, stone-lined plazas in Taino settlements across Puerto Rico used for the batu ball game, dances, and community meetings. These courts were vital for conflict resolution, social interaction, and religious rituals. In Cuentra Connmigo, the batey manifests in a non-vegetated and clean-swept yard separate from the garden plots.

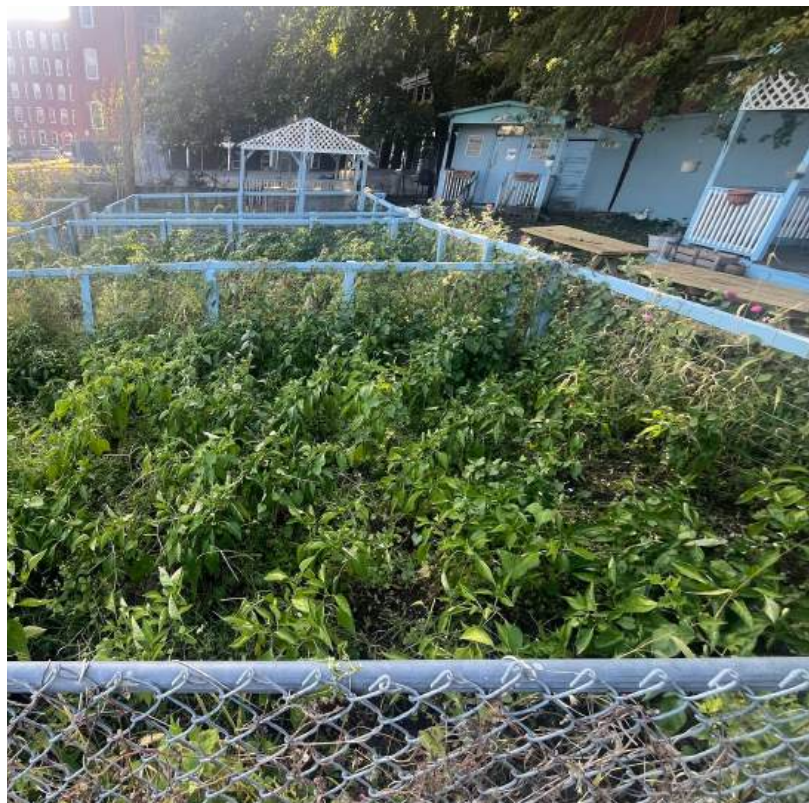


Figure 21: Cuenta Conmigo side view

The casita is always open and is recreation-oriented, and is a call to the Puerto Rican countryside. Casitas are typically painted in bright colors, the way they did in the 1930s in Puerto Rico. In Puerto Rico, casitas were also used as refuge from the hot weather as it is open. Casitas represent a vernacular architecture distinct from modern concrete structures. Usually they were constructed out of scrap wood. They typically have pitched roofs, with an entrance located at the gable end. Foundations were constructed with bricks or concrete, leading to a veranda or porch distinguished by x's in the railing or hanging plants above. In Cuenta Conmigo, we see this with the white lattice design. It is prominently situated in the garden in the rear and aligned with an entry path to the secondary gate. With this secondary gate's connection to Youth Garden, it promotes movement through these spaces.



Figure 22: Cuenta Conmigo side entrance, contiguous with Youth Garden

These features demonstrate high performance against many municipal goals: it reclaims underutilized land; provides spaces for gathering, ritual, and expressive landscapes; functions as a multifunctional open space with both productive and social areas; and enhances connectivity. The main gap here is with the initial planning "importance": the depth of cultural agency and communal building practices observed exceeds what is typically prioritized in top-down municipal planning documents, which often focus on metrics like food yield or plot numbers rather than the lived processes of cultural reproduction, vernacular placemaking, and community sovereignty that this garden embodies. The casita and batey are not just "features" —they are material testaments to user-driven spatial agency, revealing a form of resilience rooted in collective cultural practice rather than institutional design.

B. Comparative Cases

1. Gardening the Community

Gardening the Community (GtC), a BIPOC-led organization founded in 2002, represents an explicitly justice-oriented, land reclamation model. It emerged as a direct response to Springfield's severe vacancy crisis and municipal neglect, actively targeting blighted, privately-owned vacant lots in neighborhoods like Old Hill and Six Corners, characterized by

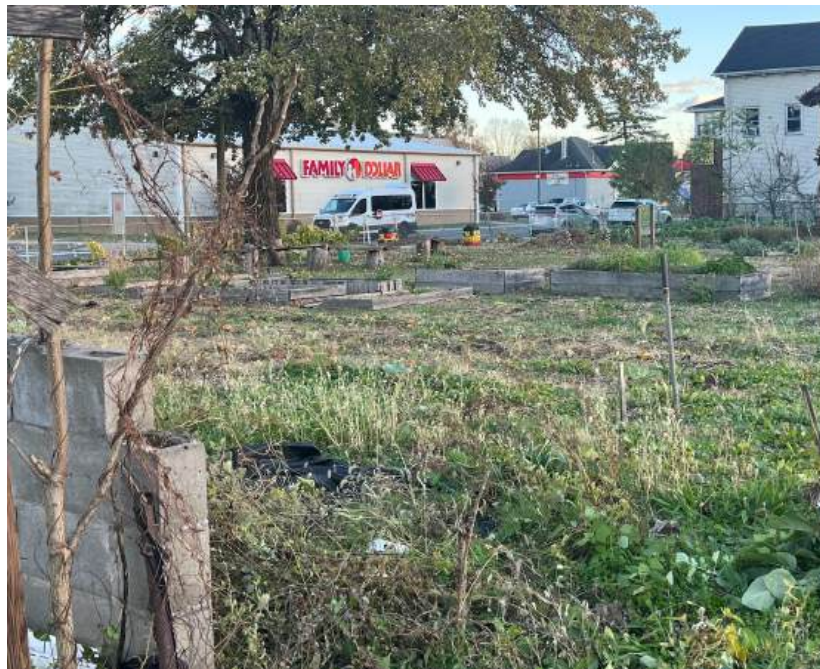
high poverty and a lack of grocery stores. GtC's mission is fundamentally activist, focusing on youth leadership, food sovereignty, and transforming landscapes of abandonment into community-controlled assets.

GtC's spatial strategy in Holyoke is defined by a strategic hub-and-spoke model, centered on its flagship Hancock Street Farm. The hub-and-spoke model is a logistical and organizational structure where a central core (the hub) coordinates activities and distributes resources to a network of satellite locations (the spokes) (Jennette, 2010). Established as the organization's primary production and educational hub in the city, Hancock Street Farm functions as the network's engine. It is a larger-scale, cohesive community garden, distinct from some of the dispersed family plots of Nuestras Raíces, and serves as the main "living classroom" for GtC's paid youth internship program. The harvest from this central site supplies a deliberately cultivated network of downstream nodes: mobile farm stands operated by youth at strategic community locations and smaller satellite gardens on reclaimed lots.

This operational model directly influences the sites' tenure and physicality. The architecture and infrastructure at GtC sites, including Hancock Street as I will show in my field observations, prioritize adaptability and mobility featuring movable hoop houses, rainwater storage, and compost systems that assert permanence through long-term cultivation rather than permanent structures. Together, these cases allow this study to probe whether the latent capacity for crisis response is better cultivated through long-term cultural integration and institutional partnership (seen predominantly in Nuestras Raíces) or through long-term infrastructural decisions (seen uniquely in GtC). Both organizations operate within the overarching failure of the conventional food system, making them prime sites to investigate the transition from food security provision to food sovereignty practice. The following analysis therefore draws on site observations conducted at Hancock Street Farm during the same timeframe as the Nuestras Raíces garden visits.



Figure 23: GtC view from the front entrance⁴



⁴ Notable features of the sign include the land acknowledgement to Mitchell Machine (the neighboring equipment/machine manufacturing company). Interestingly, there is orange spray paint covering the bottom text, "Gardening the Community is a project of the Northeast Organic Farming Association Massachusetts Chapter." Below this main sign is a handmade/handpainted sign.

Figure 24: GtC view from the back entrance

Hancock Street Farm occupies a former vacant lot in Springfield's Old Hill neighborhood – an area that, like South Holyoke, bears the marks of disinvestment, vacancy, and high food insecurity. Unlike the Nuestras Raíces gardens in Holyoke, which are often tucked behind hedges or obscured by parked cars, Hancock Street Farm presents as unmistakably public. The site is bounded by wild vegetation that serves as a permeable edge, signaling openness rather than exclusion. A Family Dollar sits directly adjacent to the farm, creating a spatial juxtaposition that speaks to the neighborhood's food landscape: a chain convenience store selling processed goods alongside a community-run farm offering fresh produce.

The most striking feature of Hancock Street Farm is its sheer productivity and diversity of crops. Unlike some Nuestras Raíces gardens where vegetable production is limited, under the same seasonal constraints, Hancock Street Farm contains a wide array of food crops that reflect actual cooking practices: yellow squash scattered across beds, heirloom tomato varieties including San Marzano, eggplant, cauliflower, herbs like curly parsley, dill, and chives, and a prepared garlic bed with planted cloves. The presence of a freshly tilled bed for garlic, planted in fall for the following year's harvest, indicates not only active cultivation but forward-looking stewardship, the kind of planning that anticipates future seasons rather than merely responding to the present.



Figure 25: Patch of a variety of tomatoes being grown atop a pile of mulch



Figure 26: Author's identification of possible tomato varieties



Figure 27: Freshly prepared garlic bed⁵

⁵ Here is a plot being prepared for planting, as shown by the long, dark strip of freshly tilled or worked soil. This appears to be a prepared garlic bed with planted cloves. Decent maintenance has been done to prevent overgrown beds.



Figure 28: Garden bed with herbs⁶

⁶ This is a close-up of a raised garden bed with a variety of green foliage. The plant in the foreground resembles curly parsley and the plant in the background is more feathery & finely-leaved, resembling dill or fennel. In the middle, there also seems to be chives or wild garlic/onion.

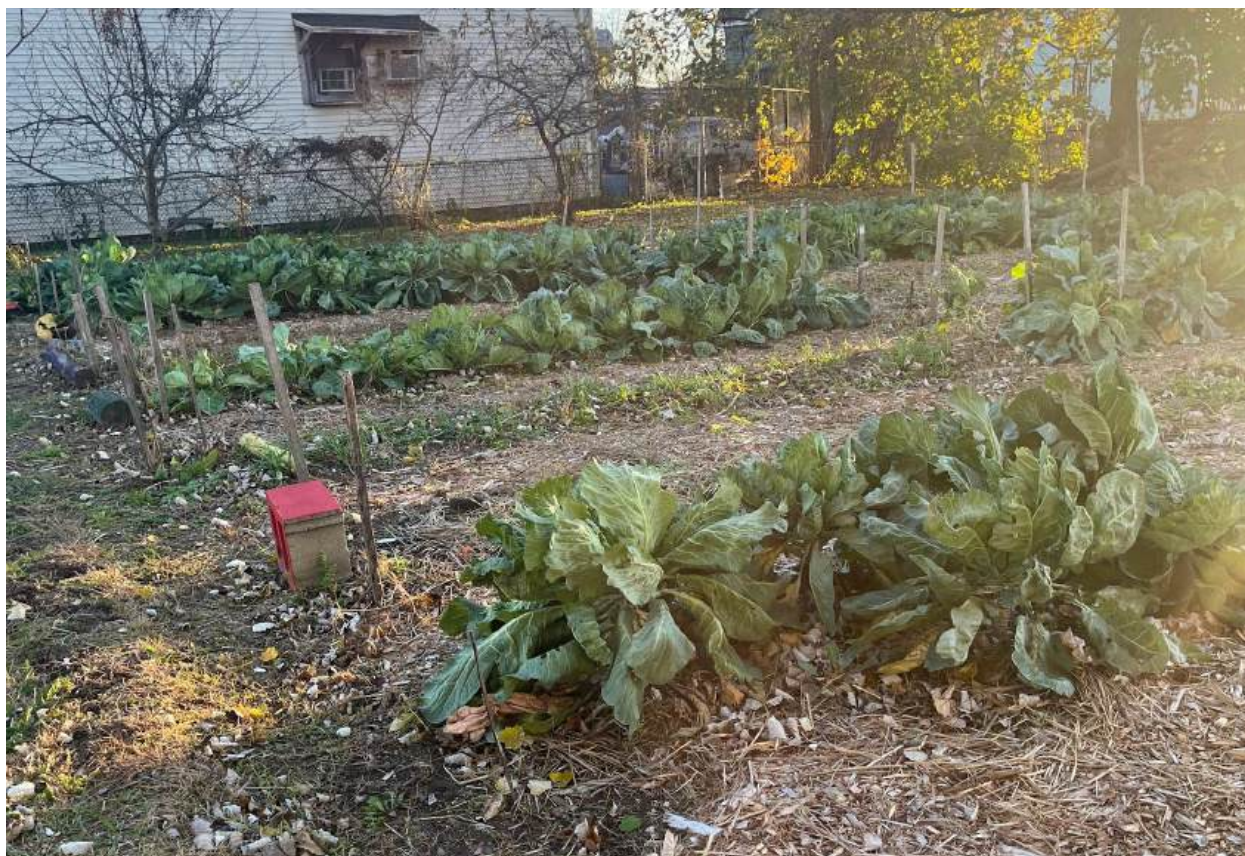


Figure 29: Active rows of cauliflower in the left side of garden

The infrastructure at Hancock Street Farm is notable for its combination of permanence and mobility. A plastic-covered hoop house extends the growing season, protecting crops from early frost and enabling year-round production. Rainwater harvesting systems – barrels connected to downspouts on the shed – demonstrate a self-reliant approach to irrigation less dependent on municipal infrastructure. A solar panel visible near the hoop house suggests attention to off-grid energy needs, a form of infrastructural redundancy that becomes valuable during system shocks. Yet these permanent elements coexist with mobile structures such as movable hoop houses, compost systems, and farm stands that can be deployed where needed. This hybrid model of permanent infrastructure for production and mobile infrastructure for distribution embodies a resilience logic that differs from both the fixed, institutionally managed gardens of the HHA and the individually claimed plots of *Nuestras Raíces*.



Figure 30: Side view of the plastic covered hoop house & adjacent solar panel at the back GtC entrance⁷

⁷ A hoop house is a cost-effective greenhouse structure, also called a high tunnel, made with a series of steel/PVC hoops. It's main purpose is to extend the growing season by retaining heat and protecting plants from weather, pests, etc.plastic-covered

The social infrastructure of Hancock Street Farm is equally deliberate. A covered porch attached to a colorful shed painted in red, green, and yellow provides shaded gathering space. A sign on the shed reads "Growing Food, Leadership & Community," framing the farm's mission in terms of youth development and collective agency. The inside of the shed is occupied by a tall stack of large, brown paper bags, filled with organic material like leaves and mulch. On each corner of the shed, we see large barrels mentioned earlier. White downspouts seem to lead from the roof into the barrels, suggesting a rainwater harvesting system. Looking at the background, the overall impression is that the area immediately surrounding and behind the main shed is significantly overgrown with wild vegetation, indicating a deliberate allowance for the plants to grow unchecked.



Figure 31: Hancock Street Farm shed

Benches made from cut logs offer seating throughout the site. A table and scattered seating, such as a chair, a stool, and a log, suggest an environment where people stop, gather, and converse. The heterogeneity of the seating –the presence of an actual chair beside an ad-hoc stool speaks – to the improvisational character of the congregation here. No delineated seating area was designed; instead, users have assembled whatever is chair-adjacent, claiming space through

the simple act of pulling something over to sit. The repetition of red, green, and yellow throughout the site in painted posts, stacked tires, and the shed's facade creates visual coherence and a sense of intentional design.



Figure 32: Array of seating at GtC's front entrance

2. Holyoke Food & Equity Collective (Holyoke Collective)

Holyoke Food & Equity Collective, known locally as Holyoke Collective, represents a different organizational logic than the territory-based model of Nuestras Raíces. Founded in 2020 amid the intersecting crises of the COVID-19 pandemic, the racial justice uprisings, and the

deepening precarity of the conventional food system, the organization emerged from a group of Holyoke residents who sought "to do it a different way," as a founding member described in my interview. More than half of the founding members were themselves residents of Holyoke, and the organization was built from the outset around principles of mutual aid, horizontal governance, and responsiveness to community-identified needs rather than pre-defined programming.

Unlike *Nuestras Raíces*, which has cultivated a network of discrete, physically defined garden sites over three decades, Holyoke Collective operates as a remote organization that doesn't own any land. As the founding member explained, "We don't own any land. We're kind of like a remote organization or play space. So we partner with other organizations that have land basically." Despite having no garden of their own, they build garden beds in residents' backyards, run a community compost hub, organize collective dinners, and coordinate a seedling share. Their commons is a distributed network of practices that weaves through private and institutional spaces across the city. Its temporality is fundamentally event-based rather than tenure-bas

ed, defined by punctuated moments of collective gathering: twice-monthly diaper distributions, four dinners a year, and seedling distributions timed to the growing season. The same Holyoke Collective member describes these gatherings as a form of "real time movement building."

The origins of Holyoke Collective offer insight into its institutional logic. "More than half of us that formed the collective, including me," the member noted, "wanted to do it a different way [than *Nuestras Raíces*]." Their response to the 2025 Healthy Incentives Program (HIP) cuts illustrates this model in action. When the state reduced HIP benefits from \$40-60 to just \$20 per person, Holyoke Collective adapted its seedling share. According to the member interviewed, they created a \$60 "solidarity option" for credit card payers to subsidize HIP recipients, ensuring that vulnerable community members could still access seedlings without dipping into their SNAP funds.

CHAPTER SIX: DIAGNOSING LATENT CAPACITY

This chapter synthesizes the planning analysis from Chapter Four and the field findings from Chapter Five to answer the central question of this thesis: why do some community gardens become vital during crises while others do not? I first diagnose the mechanisms that produce Lively, Alive, and Neglected outcomes by examining how spatial, political, and cultural agency are enacted or suppressed in Holyoke's gardens. I then turn to two comparative tensions revealed by the field, which are cultural integration versus infrastructural investment and territory-based versus practice-based commoning. Finally, I highlight three principles for cultivating latent capacity in community food infrastructure. This chapter moves from diagnosis to design: from identifying why gardens fail or thrive to proposing how planners and communities can act on those findings.

A. Garden Classifications

To understand the divergent fates of these spaces, this analysis returns to the triad of agency – spatial, political, cultural – established in the literature. My interview data reveals how these forms of agency are enacted, negotiated, or suppressed in Holyoke's gardens, directly shaping their capacity for resilience.

Cuenta Conmigo's observed "Lively" state – marked by its permanent social infrastructure (lechonera, stage) and active, diverse plots – is an outcome of integrated facilitators of spatial, political, and cultural agency. The director's account of the history of the site reveals its foundation in bottom-up political mobilization: "residents were the driving force... they identified that land... and worked with us to get permissions." This initial political act secured its spatial foothold. Its subsequent cultural performance, hosting pre-pandemic harvest festivals that the director describes as vital for "community gathering and organizing," generated further political capital. This capital was then strategically leveraged in a negotiation with the city, successfully advocating for the creation of the adjacent Pina Park. Here, the garden functioned not just as a food source but as proof of concept for community stewardship, translating cultural vitality into tangible public investment. The coordinator's emphasis on the garden's culturally significant crops, like ají dulce, underscores how its cultural agency fuels this cycle. Cuenta

Connmigo thus operates as a resilient hub because its spatial claim is continually validated and enriched by active political and cultural practice.

The "Alive" state of Nuestras Raíces' original, plot-rental gardens reflects a secure but more individualized expression of agency. The director's account of La Finquita's origin, a negotiation between migrant farmers and Providence Ministries, highlights a classic act of spatial reclamation on "borrowed" land. La Piedra's story, as the director details, shows an evolution: from a lease with Hammerman Hardware to an outright purchase using settlement funds, representing a monumental consolidation of permanent spatial agency. This security is the bedrock of their functionality. The coordinator confirms that these sites successfully perform the core municipal goal of "promoting local food systems" through the \$30/season plot model. However, their performance leans heavily towards individualized cultural agency (family plots growing cultural crops) rather than the collective, ritual-based agency seen at Cuenta Connmigo. The director notes the challenge of reactivating broader community engagement and events post-pandemic. These gardens are thus resilient in terms of tenure and household food security, but their latent potential for broader social infrastructure and political mobilization remains under-activated, placing them in the "Alive" but not fully "Lively" category.

The "Neglected" state of the Holyoke Housing Authority (HHA) gardens is a case of outsourced governance impacting the grassroots nature of the nonprofit. The coordinator's testimony identifies the mechanism that their viability is entirely dependent on the "resident coordinator" employed by the HHA. "If the person in that position wants the garden, it's way easier... but if they're not that excited about it, it's hard." At Jarvis Heights, pandemic-era turnover in this role led to disengagement where "the new one... wasn't very enthusiastic about the garden. And... it kind of lost momentum." The spatial agency for these gardens pre- and post-crisis seemed to *not* be held by the gardeners; it was contingent on the interest of a third-party institutional actor. The HHA's planning logic, focused on "low operational burden" and "amenity value," created a managerial model that stripped the community of direct control. When the human link in that bureaucratic chain dissolved, there was no grassroots political agency or deep cultural practice robust enough to sustain the space. The gardens became

"unused, peripheral spaces" because their governance model systematically prevented the cultivation of the very communal ownership and cultural investment needed for resilience.

Hidden Treasures in Beaudoin Village represents a hybrid case, illuminating the fragility of cross-sector partnerships. The coordinator describes its dual structure: half for a school garden club, half for resident plots. Its post-pandemic decline mirrors Jarvis Heights, tied to a change in the HHA coordinator. However, the director's involvement in restarting the garden club with UMass and Sullivan School faculty points to ongoing efforts to rebuild its educational and political agency. This garden's fate is tied to the stability of not one but multiple institutional partnerships (HHA, School District, Nuestras Raíces). Its performance is therefore a barometer of inter-organizational coordination. While it holds the planned "important feature" of educational space, its reliance on several precarious institutional relationships makes its social infrastructure highly vulnerable to disruption, demonstrating how even well-intentioned partnerships can create complex webs of dependency that threaten a garden's adaptive capacity.

These interview accounts demonstrate that the "performance" gap identified in observations is directly attributable to the presence or absence of integrated agency. Gardens thrive (Cuenta Conmigo) where community-driven political action secures space for deep cultural practice. They persist but remain limited (La Finquita) where spatial security is achieved but broader social-political agency is not cultivated. They fail (Jarvis Heights) where agency is fully outsourced to external institutions, leaving no community-owned social infrastructure to sustain the space during shocks.

While the preceding analysis examines how agency succeeds or fails in gardens shaped by inter-organizational cooperation (e.g., with the HHA, schools, and the city), a parallel investigation must consider the dynamics of intra-organizational cooperation within Nuestras Raíces' own network. The grower's experience at La Finca provides a high-stakes lens on the dynamics of agency within Nuestras Raíces' network. While the community gardens are framed as spaces for personal use and cultural practice, La Finca operates under a parallel but intensified logic: entrepreneurship and livelihood. This commercial dimension raises the stakes of spatial

control and social belonging, revealing in an intensified version of the latent tensions that may exist across all communal growing sites.

In the community gardens, as far as planning goes, spatial agency is negotiated with institutions (HHA, the City). At La Finca, the grower reveals that, in practice, the primary negotiation is with other plot-holders. The grower's description of veteran farmers who "rooted themselves so much into that land where they feel it's not the public land, it's their land" points to an informal privatization of communal space. When a plot is one's business or familial asset, the instinct to defend territory intensifies. The spatial agency goes beyond claiming land from the city and is about claiming exclusive control within the claimed land—a micro-scale enclosure. While the director described the organization's external "political" strategy as limited by its apolitical 501(c)(3) status and dependent on back-end capacity, the grower exposes an internal political landscape. Power is not just held by institutions, but by a dominant, entrenched social group within the growing space. The grower describes a political agency wielded as exclusion: "they all get together and they create these situations" to "kick out" newer farmers. This is a political agency turned inward, used to police the boundaries of the community rather than to defend it from external threats.

The grower also affirms the profound cultural and therapeutic value of the space: it "brought back my faith in humanity" and provided a sense of purpose. This aligns with the literature on cultural agency as spiritual nourishment. However, the grower also identifies the dark side where the very strength of the long-standing cultural in-group creates a barrier to entry. The cultural knowledge and practices are not freely transferred but guarded. When the grower says, "they're going to take it with them," they diagnose a failure of the intergenerational knowledge transfer that is central to cultural agency's definition. The culture becomes static and proprietary instead of living and shared.

The grower's testimony at La Finca acts as a stress test for the social model used in Nuestras Raíces' gardens. It reveals that the "rent-a-plot" system, whether for \$30/season or for a business, can inadvertently foster individualized territoriality over collective stewardship if not actively managed. Long-term tenure without explicit community governance mechanisms can

lead to the formation of insular power blocs that resist renewal and inclusion. The cultural bonds that create profound resilience for existing members can become exclusionary boundaries for newcomers, threatening the organization's long-term vitality and adaptive capacity.

Thus, the challenges the grower faces are not anomalies of the farm program; they are the amplified symptoms of systemic risks present in any long-term, plot-based communal space. The grower's insights provide a warning that designing for resilience requires not only securing land from external threats but also designing internal social and governance structures that proactively combat insularity and guarantee equitable access to the space, its knowledge, and its benefits.

B. Comparative Tensions

The analysis above reveals how agency operates within Nuestras Raíces gardens. But comparing across organizations reveals additional dimensions of resilience. I now turn to two tensions that emerged from comparing Nuestras Raíces with Gardening the Community and Holyoke Collective.

Is there a difference between prioritizing cultural integration versus infrastructural investment?

Nuestras Raíces's model of resilience is built on a goal toward cultural integration: thirty years of bicultural practice, intergenerational knowledge transfer, and political negotiation have embedded its gardens in the social fabric of Holyoke's Puerto Rican community. This is resilience as a function of belonging –the kind of deep attachment that kept La Finquita cultivated through institutional shifts, that enabled Cuenta Conmigo to translate cultural vitality into public investment, and that made these spaces meaningful beyond their productive yield. GtC's model, by contrast, is built on infrastructural investment such as hoop houses that extend the growing season, rainwater harvesting that reduces dependence on municipal systems, solar panels that provide off-grid energy, and farm stands that distribute produce through multiple channels. This is resilience as a function of redundancy –the kind of material buffer that enables a site to function when water systems fail and when the grid goes down.

The tension here is between two different theories of what makes a space resilient in crisis. Is it the depth of social ties, like the networks of mutual aid, cultural practice, and collective governance, that can be mobilized when needed? Or is it the material infrastructure,

like the technologies that reduce dependency on fragile systems and create redundancy? The Nuestras Raíces gardens have the former but unevenly distributed; Cuenta Conmigo has it in abundance, while La Finquita and La Piedra have it in more attenuated form. GtC has the latter in abundance, but whether it has the former – that is, the deep social embeddedness that sustained Cuenta Conmigo through the pandemic – is less visible from the material traces alone.

From the site observations from GtC's Hancock Street Farm, the site would be classified as "Lively" like Cuenta Conmigo. It exhibits high adaptive social use, strong visible networks, and evidence of collective life beyond individual cultivation. The presence of communal gathering spaces, youth programming infrastructure, and a distribution network that moves produce from farm to community distinguishes it from the individually oriented "Alive" gardens in the Nuestras Raíces network. However, the sources of this liveliness differ from Cuenta Conmigo. Where Cuenta Conmigo's liveliness emerged from bottom-up political mobilization and deep cultural embedding, Hancock Street Farm's liveliness stems from an organizational model explicitly designed around accessibility, youth leadership, and productive distribution.

The question, then, is whether these forms of resilience are substitutes or complements. Can cultural integration substitute for infrastructural investment? Can infrastructural investment substitute for social embeddedness? Or does the most resilient model require both?

Is there a difference between the "commons" emerging from shared territories and the commons emerging from shared practices?

The distinction in spatial logic between findings Nuestras Raíces and Holyoke Collective from Chapter Five reveal two distinct modes, and potentially complementary, modes of commoning.

Nuestras Raíces leans toward a territory-based commons. Its model is rooted in the acquisition and stewardship of specific parcels –from the foundational La Finquita, established on a former parking lot, to La Piedra, whose outright purchase in 2023 secured permanent community control. This territorial focus enables the construction of permanent physical infrastructure: the casitas, lechoneras, and stages that anchor collective life and enable deep cultural embedding. The temporal rhythm of this commons follows a logic of tenure and annual

return, structured by seasonal plot rentals (\$30/season) and punctuated by annual harvest festivals. However, as observed in the post-pandemic disengagement at some Nuestras Raíces sites, this territory-based model can be vulnerable when the habitual repetitions of daily life, the "rhythms" of commoning, are disrupted.

In stark contrast, Holyoke Collective embodies a practice-based commons that operates independently of land ownership. Their distributed network of backyard beds, compost hubs, and collective dinners weaves through private and institutional spaces across the city. Its temporality is fundamentally event-based rather than tenure-based, defined by punctuated moments of collective gathering. The same Holyoke Collective member describes these gatherings as a form of "real time movement building"—a rhythm of commoning that may be inherently more resilient to post-pandemic disengagement, as it does not depend on year-round, individual labor or plot maintenance.

This comparison illuminates a productive ambiguity in how private property participates in urban commons. Holyoke Collective's practice of building garden beds in backyards actively transforms private property—those interstitial spaces situated between homes, and between home and street—into nodes of a distributed commons network. When a Holyoke Collective member describes this model as giving residents "more like food sovereignty or or not necessarily food sovereignty, but like more ownership right over their gardens," her hesitation is revealing. It points to a hybrid condition where these are private yards mobilized for collective food resilience through shared practice. The garden bed is privately located but collectively initiated, supported by shared seedlings from the Collective's community supported agriculture (CSA), maintained with knowledge exchanged through workshops, and integrated into a citywide mutual aid infrastructure. Here, private property does not preclude commoning.

This approach stands in contrast to the Holyoke Housing Authority model examined earlier. Where the HHA gardens positioned residents as tenants of institutional amenities—their access and stewardship contingent on the enthusiasm of a single coordinator—Holyoke Collective's backyard model positions residents as co-creators of a distributed commons. The HHA approach outsources spatial agency to an institutional gatekeeper; the Collective's approach embeds agency in the hands of residents themselves, supported by a network of shared resources

and reciprocal relationships rather than a bureaucratic chain of command. Applied to the residential gardens of West Holyoke, this alternative logic suggests a trajectory where, instead of a centralized garden whose fate hinges on institutional continuity, a distributed network of backyard beds, collectively supported, might prove more resilient to the shocks like coordinator turnover and funding cuts that left El Renacer and Hidden Treasure vulnerable.

The origins of Holyoke Collective reveal a perceived trade-off in institutionalization. The Nuestras Raíces model offers the undeniable strengths of security, longevity, and permanent infrastructure, but it also entails bureaucracy, dependency on municipal and housing authority partnerships, and the vulnerability documented at sites like Jarvis Heights, where a garden's viability hinged largely on the enthusiasm of a single HHA-employed coordinator. Holyoke Collective's "bare bones" model—a tiny budget, "very part time," operating remotely as its member described—purchases flexibility and autonomy at the cost of limited scale.

These divergent models enable different kinds of resilience. Territory-based commons, as exemplified by Cuenta Conmigo, enable deep cultural embedding and the accumulation of permanent assets, infrastructure that can shelter a harvest festival today or serve as a distribution node in a crisis tomorrow. Practice-based commons, as embodied by Holyoke Collective, enable rapid adaptation and penetration into spaces institutional gardens cannot reach. Referencing back to the Collective's response to the 2025 Healthy Incentives Program (HIP) cuts, they exhibited mutual aid infrastructure in action made possible precisely by their non-territorial, practice-based model.

This analysis suggests that a resilient food system cannot afford to choose between these models. The permanence of territory-based commons and the adaptability of practice-based commons are not mutually exclusive but deeply complementary. The former roots communities in place, enabling the slow accumulation of cultural knowledge and political capital. The latter weaves those places into a flexible, responsive network capable of improvisation when shocks arrive. The most robust urban commons may be those that, like the relationship between Nuestras Raíces and Holyoke Collective, exist in productive tensions of territory and practice, & permanence and flexibility (see Table 3).

Table 3: Comparative Dimensions of Territory-Based and Practice-Based Commons

Dimension	Nuestras Raíces (Territory-Based)	Holyoke Collective (Practice-Based)
Spatial logic	Discrete parcels; permanent infrastructure	Distributed; private yards; partner sites
Temporal rhythm	Seasonal plots; annual festivals	Event-based; crisis-responsive
Property relation	Leased/purchased land; institutional partnerships	Private yards with garden beds; no land ownership
Governance	Formal nonprofit with staff	Volunteer collective; "bare bones"
Crisis response	Vulnerable to coordinator turnover; post-pandemic disruption	Rapid adaptation (HIP response); flexible programming
Agency cultivated	Spatial & cultural (individual/collective mix)	Political & mutual aid (collective practice)

Overall, these comparative tensions reveal that resilience is not a single property but a set of trade-offs. The question, then, is how design can navigate these trade-offs. I now extract three principles for designing gardens capable of weathering crises.

C. Implications for Crisis Response

The divergence between gardens that become vital during crisis and those that fade is a matter of design. This investigation reveals that resilient response is an emergent property of specific, cultivable socio-spatial conditions. Understanding these conditions requires examining how design decisions from the scale of entry points to the configuration of plantings shape a garden's capacity to absorb shock and mobilize networks.

1. Principle 1: Design for legible boundaries of responsibility

Do scale and entry matter?

The gates and locks that perimeter many community gardens present an apparent paradox. While they may give the impression of severely limited public access, these boundaries often protect a delicate "carrying capacity" that could be easily destroyed by unlimited use. Local initiators have decided that their community open-space projects can provide no more than limited physical access. Yet almost all are designed to have unlimited public visual appeal. This tension points to a deeper design question: how does a space balance accessibility with the stewardship intensity required for its survival? The most resilient gardens in Holyoke are not necessarily the most open, but those where the logic of entry aligns with the logic of care.

Cuenta Conmigo's secondary gate connects directly to the Youth Garden and Pina Park, creating multiple points of entry that invite movement through the space without demanding anything of the passerby. Yet its primary entrance is clearly demarcated by blue picket fences. These fences are legible in the sense that a newcomer can enter, look around, and leave without obligation. But it also signals what the garden asks of those who stay: the *lechonera* implies shared cooking, the *casita* implies gathering, and the cultivated plots imply stewardship. At the opposite extreme, El Renacer's padlocked gate and car-obscured entrance signaled exclusion. A passerby could not tell whether the garden was abandoned or private, cared for or forgotten. The HHA gardens failed because their boundaries offered no legibility about the relationship between access and responsibility. Between these opposite ends of the spectrum, the "Alive" gardens of La Finquita and La Piedra present a different tension. Their gates are accessible, their plots are rented, but there is no communal infrastructure that asks anything of users beyond maintaining their individual plot. The logic of entry, being "open" in these instances, does not align with a logic of care, being individual and plot-oriented. The result is functionality without the latent capacity for collective crisis response that Cuenta Conmigo's legible boundaries enable.

Further, the scales at which the gardens are bounded matters. In the case of rented plots, there are lasting biophysical and social links between the plots and common life of the city block/neighborhood landscape. A garden with no formal or individual plots is spontaneous, but also relies on the expertise of the grower for what, where, and when to grow. La Finquita's 32 individual plots are each bounded at the scale of a family. A renter arrives, tends their designated soil, harvests their designated crops, and leaves. The biophysical link between plot and grower is

strong and direct. But the social link between plot and neighborhood is weak. The garden's hedges obscure it from the street. There is no communal seating, no shared cooking infrastructure, no space that asks anything of users beyond their individual plot. The result is a commons that functions as an aggregate of private territories. When crisis strikes, these individual plot-holders have no collective infrastructure to activate. Cuenta Conmigo also has rental plots, approximately 15 to 20 of them, with the broader garden bounded at multiple scales simultaneously. The individual plot is the smallest unit of care. Holyoke Collective inverts the logic entirely. There are no garden boundaries because there is no garden. Instead, individual backyards become nodes in a distributed network. While the scale of bounding is the household, the logic of commoning operates across households through shared seedlings, collective dinners, and mutual aid.

If we treat the commons as discrete territories, livelihoods, and resource management largely distinct from private property, logics of private property are still intertwined with logics of commoning. What these comparisons reveal is that bounding should be about the scale of responsibility. A garden bounded only at the plot scale produces individual food security but not collective crisis response. A garden bounded at multiple scales like the plot, garden, and neighborhood can hold individual practice and collective action in tension. A network bounded at the household scale but linked through shared practice can achieve resilience without any garden at all.

A case for delineation.

When the garden's fate is tied to the stability of not one but multiple institutional partnerships (HHA, School District, Nuestras Raíces), its performance becomes a barometer of inter-organizational coordination. While it holds the planned "important feature" of educational space, its reliance on several precarious institutional relationships makes its social infrastructure highly vulnerable to disruption, as demonstrated during the COVID-19 pandemic.

This is a case for delineation, not as exclusion, but as clarity of relationship. Spaces that thrive under stress are those where the boundaries of responsibility are legible, where users understand not only what the space offers but what it asks of them. This kind of legibility is

needed when spaces are heterotopias. "Heterotopias," as coined by Foucault, are real places capable of juxtaposing several spaces and several sites that are in themselves incompatible. Community gardens often function as heterotopias, but their capacity to hold contradiction should be designed for. The question is whether garden design features can reconcile these incompatibilities, creating spaces that are simultaneously productive and social, private and collective, and cultivated and wild.

In *Cuenta Conmigo*, we see this reconciliation achieved through multifunctional spaces such as the lechonera that blur the boundary between kitchen and garden. The lechonera is both a structure for cooking and a device that transforms harvested crops into shared meals, that draws people from their individual plots into collective celebration, and that activates the garden as a site of cultural reproduction. Similarly, the adjacent stage hosts harvest festivals and community gatherings, programming the open space not as an empty lawn but as an arena for the rituals that bind a community. This logic of multifunctionality has implications for how gardens accommodate new entrants. Policies oriented toward welcoming newcomers require visible delineation – that is, spatial cues that communicate how to enter, where to go, what is shared and what is claimed. In an environment that does not promote organic assembly, where casual passersby may not instinctively understand the garden's social logic, design must provide orientation. This might mean a gathering space visible from the street, a welcoming entrance that invites curiosity, or communal infrastructure that signals the garden's collective dimension before a newcomer ever touches a trowel as seen in GtC's Hancock Street Farm.

2. Principle 2: Design against micro-territories by designing for multifunctionality and mobility.

Risk of Micro-Territories

Conversely, design that outsources stewardship or prioritizes managerial ease inherently disables crisis capacity. Gardens subject to housing authority logic, where engagement hinges on a single staff coordinator, are designed for consumption, not ownership. Their spatial agency is leased. As seen in *El Renacer* in Jarvis Heights, the garden had the form of a community garden with fencing, signage, and a shed, but lacked the ongoing stewardship that would activate it as a living space. The spatial agency was contingent on a third-party institutional actor whose interest

could not be guaranteed. Hidden Treasure at Beaudoin Village told a similar story. Its decline also followed a change in HHA coordinator. But here, the evidence was even more stark with a discarded half-Puerto Rican, half-American flag on the fence – a symbolic artifact of a partnership that had eroded. Further, the student side and resident side bled into one another without demarcation. This garden planned as a shared community asset between the Housing Authority, the school district, and Nuestras Raíces performed as an unclaimed liminal zone with no architectural identity or invitation.

This design feature alluding to the existence of an institutional gatekeeper, systematically prevents the formation of the grassroots political networks and deep cultural investment required for adaptive response. When the coordinator leaves, the social circuit is broken. This means design must address the internal social architecture of the space. The rent-a-plot model, while providing clear individual tenure, carries the latent risk of fostering insularity. As seen at the broader urban farm La Finca, spatial agency can involve both a negotiation with institutions and other plot-holders. A garden can become a patchwork of fortified micro-territories, its social infrastructure incapable of the unified, expansive action a crisis demands.

The design feature that counters this is more procedural than physical, suggesting the importance of governance mechanisms that actively cultivate bridging social capital and intergenerational knowledge transfer. Plant exchange, for instance, has a seasonal rhythm that can integrate newcomers into the garden's social life. The persistence of certain perennial varieties across plots suggests sharing and cross-pollination of knowledge. Different plants enable different relations, and the homogeneity of plants across plots can indicate an informal economy of cuttings and divisions. Sharing perennials helps newcomers establish themselves, connecting them to longer-term growers through the material practice of propagation. But this only occurs when there is a cycle of inhabitants: when experienced growers remain present to share and when governance structures encourage them to do so.

In GtC's Hancock Street Farm, there is no rent-a-plot system, no designated family beds, and no individual territories to defend. This collective governance creates certain conditions that individual-plot models cannot replicate. The combination of annuals and biennials across a

single, unified site creates temporal continuity that bridges growing seasons. The presence of a freshly tilled garlic bed, a biennial planted in fall for the following year's harvest, alongside annuals like heirloom tomatoes, yellow squash, and eggplant, signals a stewardship horizon that extends beyond any single season. There is no moment when the garden is "empty" or between caretakers because the plantings themselves encode anticipation. A garlic bed planted in October is a promise made to the following July. This temporal depth is difficult to achieve in a rent-a-plot model, where individual growers may or may not return season to season. Moreover, the scattered organic growth, seen in the permeable edges where cultivated space meets unmanaged space, creates ecological redundancy. Unlike the manicured boundaries of La Finquita or the sterile edges of El Renacer, GtC's wild margins are deliberate. The overgrown areas provide habitat for pollinators, retain soil moisture, and create microclimates that buffer against temperature extremes.

A concern across many community gardens is that primary participants are over sixty and semi-retired. The physical capacities to keep up with gardening tasks increasingly inform how and how often these inhabitants can cultivate, both in the community garden and in their own yards. At the same time, their years of experience enable a kind of ease and skill that shapes and responds to the plants and space in ways younger growers cannot replicate. Commoning entails embodied labor, and commoners' abilities to engage in such labor change over time as communities age. Design that anticipates this transition –through accessible bed heights, reduced maintenance requirements, or mentorship structures that pair experienced growers with younger apprentices– can sustain the garden's social infrastructure across generational shifts.

Falcetti Towers, the senior-focused garden in Holyoke, demonstrates what design that anticipates this transition looks like in practice. The raised beds are fenced with wire barely visible from afar – a design prioritizing barrier-free access over visual enclosure. The garden is low-intensity, focused on easy-to-maintain flowers, herbs, and sensory plants rather than intensive food production. Benches are shared with the adjacent park, integrating rest into the experience of cultivation. The design matches the physical capacities of its users: accessible bed heights, reduced maintenance requirements, seating distributed throughout the space.

3. Principle 3: Design for complementary modes of commoning.

The Plant's-Eye View

The spatial scale of planting is both a landscape architecture concern and a social infrastructure decision. The size and scale of planting affect the scale of public space, shaping how people encounter each other and the garden. Cuenta Conmigo demonstrates this principle in action. The garden's perimeter is defined by vegetation at varying heights. Low ground covers near the entrance invite intimate engagement. Taller plantings along the edges create a partial enclosure, offering privacy for the casita and lechonera without fully blocking views from the street. The adjacent Pina Park provides open space where planting scales are below eye level, inviting different social behaviors: running, gathering in groups, and unstructured play. However, the garden's secondary gate connecting to the Youth Garden has high vegetation along this pathway, blocking sightlines between the gardens and between the Youth Garden and park specifically. This can explain why the Youth Garden's beds are sparse and neglected. Without visual connection to Cuenta Conmigo's and Pina Park's active spaces, the Youth Garden becomes invisible.

Small shrubs and ground covers provide rich ground space with varied colors and textures, inviting intimate engagement and close looking. Taller plantings create enclosure, blocking views and providing privacy. This dichotomy is seen in Cuenta Conmigo's intentional planting and El Renacer's uncontrolled growth. In El Renacer's case, the perennial weeds and volunteer plants are not designed explicitly as spatial separators, resulting in them repelling engagement and signalling lack of care instead.

Street trees act as the main medium for separating and linking gardens with surrounding buildings, extending its social reach into the neighborhood. At La Finquita, the absence of street trees along Cabot Street creates a different condition. The garden is set back from the sidewalk, its perimeter defined by hedges. There is no tree canopy extending the garden's presence into the street. The result is a garden that is easy to miss. For instance, I as a pedestrian walking past La Finquita would not have known it was there without entering the vacant parking lot. At Cuenta Conmigo, by contrast, the relationship to Pina Park extends the garden's presence into the

neighborhood. The park's trees shade the garden's edge such that the children playing in the park clearly see the garden's casita. The garden borrows the park's visibility, and the park borrows the garden's cultural specificity.

Open spaces, where planting scales are below eye level, invite different social behaviors than enclosed spaces, where vegetation blocks views and provides high degrees of privacy. GtC's Hancock Street Farm shows the effect of deliberate wild margins, as aforementioned. There is no moment where the public street ends and the private garden begins such that a passerby or Family Dollar shopper can see into the farm from the sidewalk.

This attention to the spatial logic of planting points toward a conception of plants as a medium of spatial design. But plants also operate as a medium of social design, and their social work extends beyond what can be captured in a static plan. Plants are so often seen as stationary in space, rooted and immobile. But high plant mobility within and between gardens fosters sharing, trading, and the cross-pollination of not only plant varieties but of relationships as well. As seen in Chapter Five with Gardening the Community, the presence of an adjacent farm stand creates a distributed network through which plants move from growing sites to consumption sites, connecting producers and eaters across the city. A young person who helped plant the garlic might be the one who sells it at the farm stand. The person who buys it might live three blocks away. The plant carries with it not just nutrition, but also the face of the person who grew it. This plant mobility is also social mobility, where plants carry with them the stories of their cultivation, the knowledge of their care, the relationships of their exchange.

CHAPTER SEVEN: CONCLUSION

The literature on community gardens is, in many ways, a literature of mythmaking. We tell ourselves that gardens are grassroots efforts started by neighbors and local activists, yet many rely on federal, state, and municipal programs that may have no connection to the neighborhood. We celebrate gardens as participatory assets, yet planning culture has historically treated them as interim uses rather than permanent public resources. We mobilize gardens in response to crises like the depression of 1893, the world wars, the urban disinvestment of the 1970s, yet just as repeatedly abandon them when the crisis passes. These myths shape how gardens are planned, funded, and maintained. They produce the systemic gaps that this thesis has traced: the gap between crisis-driven garden creation and post-crisis stewardship and the gap between user-initiated spaces and spaces created from institutional partnerships.

Holyoke offers a corrective to these myths. By tracing the material traces of seven gardens across a single organization, by interviewing the staff and growers who sustain them, and by comparing territory-based and practice-based models of commoning, I have documented what actually happens when gardens are planned, maintained, neglected, and, sometimes, revitalized. The findings challenge the homogenizing assumptions that structure how we think about community gardens and their role in crisis response.

Chapter Four revealed that the formal importance ascribed to community gardens in Holyoke is shaped by two parallel planning logics: city-led urban revitalization (focused on the downtown urban core) and housing authority-led affordable housing management (focused on West Holyoke and senior housing). As a result, the "importance" of garden features is institutionally fragmented.

Holyoke's 2012 Urban Renewal Plan and 2009 Center City Vision Plan frame gardens in the urban core, such as La Finquita, La Piedra, and Cuenta Conmigo, as tools for revitalization: converting blighted lots into community assets, increasing walkability and neighborhood density, serving existing high-density renter-occupied areas. The Holyoke Housing Authority's Moving-to-Work plans frame gardens at Beaudoin Village, Jarvis Heights, and Falcetti Towers within a managerial logic of risk mitigation, low operational burden, and amenity value. The 2017 Community Action Plan, by contrast, translates municipal goals into food sovereignty

action, explicitly linking Nuestras Raíces' work to city redevelopment and prioritizing culturally appropriate produce in South Holyoke, a USDA-defined food desert.

These divergent logics produce contradictions that this thesis has traced from plan to plot in Chapter Five. Gardens are simultaneously framed as revitalization catalysts in urban revitalization plans, compliance-driven amenities by housing authorities, and sovereign food systems by more community-oriented plans. While municipal plans acknowledge "community engagement," the planned features emphasize physical and managerial outcomes: yield, safety, compliance. The result is that the HHA's planning logic, focused on "low operational burden" and "amenity value," created a managerial model that stripped the community of direct control. When the human link in that bureaucratic chain dissolved, there was no grassroots political agency or deep cultural practice robust enough to sustain the space.

Yet some gardens in this study resist this fragmentation. Cuenta Conmigo demonstrates what happens when planning documents are not the only source of importance. The depth of cultural agency and communal building practices observed there exceeds what is typically prioritized in top-down municipal planning documents. The main gap at Cuenta Conmigo is not with performance but with the initial planning "importance": the planners did not anticipate a garden that would function as a heterotopia, a single real space capable of holding incompatible meanings. But Cuenta Conmigo also reveals a danger. The same spatial agency that enables cultural embedding can also be used to exclude new entrants. At La Finca, veteran farmers used their entrenched position to isolate newer farmers. The reclamation of underutilized land works in a rent-a-plot model that is not dependent on organic assembly – but that same model, without governance mechanisms to ensure turnover, can become a patchwork of fortified micro-territories. To what extent can spatial agency be used to exclude? The answer, from La Finca, is: extensively. The lesson is that designing for resilience requires not just securing land from external threats but also designing internal governance structures that guarantee equitable access for new entrants.

Planning documents encode assumptions that become material outcomes, but the inverse is also true. Gardens can encode counter-assumptions that planning documents never anticipated. The task for planners and designers is to learn from both the engineered failures of the HHA

model and the vernacular successes of Cuenta Conmigo, and from the infrastructural investment of GtC and the practice-based adaptability of Holyoke Collective.

What, then, does this thesis teach us about designing for resilience? The tensions revealed in Chapter Six –between scale and entry, delineation and openness, micro-territories and collective space, the plant's-eye view and social practice, territory-based and practice-based commons, cultural integration and infrastructural investment – point toward three principles that hold these tensions in productive balance.

Principle 1: Design for legible boundaries of responsibility

The gates and locks that perimeter many community gardens present an apparent paradox. They give the impression of severely limited public access, yet local initiators often decide that their community open-space projects can provide no more than limited physical access while maintaining unlimited public visual appeal. The question that emerged in Chapter Six of "do scale and entry matter?" reveals that the most resilient gardens are not necessarily the most open, but those where the logic of entry aligns with the logic of care.

This principle challenges the myth that gardens must be either fully open as a public good or fully closed. It challenges the assumption that permanent enclosure secures spatial agency. Instead, I suggest that resilience requires legibility in the form of spatial cues that communicate how to enter, where to go, what is shared and what is claimed. In an environment that does not promote organic assembly, where casual passersby may not instinctively understand the garden's social logic, design must provide orientation. A gathering space visible from the street, a welcoming entrance that invites curiosity, and communal infrastructure that signals the garden's collective dimension before a newcomer ever touches a trowel are the design moves that cultivate entry without sacrificing care.

Principle 2: Design against micro-territories by designing for multifunctionality and mobility.

The risk of micro-territories emerged in Chapter Six as a central threat to collective resilience. The rent-a-plot model, while providing clear individual tenure, carries the latent risk of fostering insularity. The La Finca grower's testimony revealed this in its most extreme form where veteran farmers who "rooted themselves so much into that land where they feel it's not the

public land, it's their land," who "all get together and they create these situations" to isolate newer farmers. When a plot becomes one's business or familial asset (in the case of rented plots), the instinct to defend territory intensifies. Spatial agency goes beyond claiming land from the city and becomes about claiming exclusive control within the claimed land—a micro-scale enclosure.

Cuenta Conmigo counters this risk through multifunctionality. Its lechonera is a device that transforms harvested crops into shared meals, that draws people from their individual plots into collective celebration, and that activates the garden as a site of cultural reproduction. It offers infrastructure that actively works against the fragmentation that the rent-a-plot model can produce.

Gardening the Community's Hancock Street Farm counters the risk through mobility. The plant's-eye view developed in Chapter Four revealed that plants are so often seen as stationary in space, rooted and immobile. But high plant mobility within and between gardens fosters sharing, trading, and the cross-pollination not only of varieties but of relationships. GtC's hub-and-spoke model of permanent infrastructure for production and mobile infrastructure for distribution creates a distributed network through which plants move from growing sites to consumption sites, connecting producers and eaters across the city. This plant mobility is also social mobility, where plants carry with them the stories of their cultivation, the knowledge of their care, the relationships of their exchange.

This principle challenges the myth that permanence is the only path to resilience. It challenges the assumption that individual plot tenure is the most secure form of spatial agency. Instead, I suggest that resilience requires infrastructure, whether multifunctional or mobile, that actively works against the formation of fortified micro-territories, that creates the conditions for sharing, exchange, and collective practice.

Principle 3: Design for complementary modes of commoning

The comparative analysis in Chapters Five and Six revealed a fundamental distinction between territory-based and practice-based commons. Nuestras Raíces leans toward territory-based commoning rooted in the acquisition and stewardship of specific parcels, enabling permanent physical infrastructure like the casitas, lechoneras, and stages that anchor

collective life. Holyoke Collective embodies practice-based commoning, operating independently of land ownership, building garden beds in residents' backyards, running a community compost hub, organizing collective dinners, and coordinating a seedling share. Its commons is a distributed network of practices that weaves through private and institutional spaces across the city.

The permanence of territory-based commons and the adaptability of practice-based commons can be deeply complementary. The former roots communities in place, enabling the slow accumulation of cultural knowledge and political capital. The latter weaves those places into a flexible, responsive network capable of improvisation when shocks arrive. Holyoke Collective's response to the 2025 HIP cuts, creating a \$60 solidarity option for credit card payers to subsidize HIP recipients, was mutual aid infrastructure in action, made possible precisely by its non-territorial, practice-based model. Yet this adaptability would be meaningless without the territory-based gardens that anchor cultural practice and political continuity.

A parallel tension emerged between cultural integration and infrastructural investment. Nuestras Raíces's resilience is built on cultural integration in the form of thirty years of bicultural practice, intergenerational knowledge transfer, and political negotiation. GtC's resilience is built on infrastructural investment such as hoop houses that extend the growing season, rainwater harvesting that reduces dependence on municipal systems, and solar panels that provide off-grid energy. The question raised in Chapter Six of whether these forms of resilience are substitutes or complements finds its answer in the gardens themselves. Cuenta Connigo integrates both in how its cultural practices are activated through infrastructure that enables collective life. Hancock Street Farm integrates both in how its technological infrastructure is embedded in a mission of youth leadership and community access.

This principle challenges the myth that there is a single model of resilience worth replicating. It challenges the assumption that communities must choose between permanence and flexibility, between cultural integration and infrastructural investment. Instead, I suggest that the most robust urban commons are those that exist in productive tension of territory and practice, permanence and flexibility, and cultural integration and infrastructural investment—holding these contradictions together rather than resolving them.

I return, finally, to the question that opened this thesis: what happens when the federal food support for a city suddenly vanishes? The myths that structure garden planning would have us believe that new gardens will spring up in response, that crisis creates resilience. Holyoke teaches us otherwise. The gardens that thrived under the SNAP cuts were those whose latent capacity had been cultivated over decades. They were gardens whose boundaries made legible the relationship between access and care. They were gardens whose infrastructure, whether multifunctional or mobile, worked against the fragmentation that isolates growers from collective response. They were gardens that held in productive tension the territory-based security of deep roots and the practice-based adaptability of distributed networks. They were gardens that, in their material traces and their daily practices, had already been preparing for crisis all along.

List of Tables and Figures

Table 1: Selected Nuestras Raices Gardens (p. 27–28)

Table 2: Holyoke's Municipal Planning Criteria and Corresponding "Important" Garden Features (p. 38–40)

Table 3: Comparative Dimensions of Territory-Based and Practice-Based Commons (p. 80–81)

Figure 1: (a) tract resolution of dominant race (in %) from 2020 US Census; (b) tract resolution of median household income (in \$) from 2024 American Consumer Survey; Study areas highlighted in yellow; by Justice Map (p. 23–24)

Figure 2: Chosen Nuestras Raíces Sites (p. 28)

Figure 3: URP Area Map, updated in 2020 (p. 30–31)

Figure 4: Housing type and location of garden for (a) Falcetti Towers, (b) Beaudoin Village and (c) Jarvis Heights using Google Maps images (p. 41)

Figure 5: Holyoke study area with site locations & 0.25mi "Ped Shed"; By author (p. 43)

Figure 6: Holyoke study area with Mass Land Cover 2016 Data; By author (p. 44–45)

Figure 7: Holyoke sites & neighboring social infrastructure (schools, churches, and parks); By author (p. 46–47)

Figure 8: Holyoke sites & food resources; By author (p. 48)

Figure 9: Renacer Shed (p. 49)

Figure 10: Renacer surrounding pathway (p. 50)

Figure 11: Hidden Treasure sign found adjacent to main entrance (p. 51)

Figure 12: Hidden Treasures Residential Side (p. 52)

Figure 13: Hidden Treasures Student Side (p. 52)

Figure 14: Hidden Treasures shed (p. 53)

Figure 15: La Finquita front entrance (p. 54)

Figure 16: La Finquita adjacent residential complex (p. 55)

Figure 17: La Finquita signage (p. 55–56)

Figure 18: Falcetti Towers garden beds (p. 57)

Figure 19: Cuenta Conmigo main entrance from Pina Park (p. 58)

Figure 20: Cuenta Conmigo social infrastructure (p. 59)

Figure 21: Cuenta Conmigo side view (p. 60)

Figure 22: Cuenta Conmigo side entrance, contiguous with Youth Garden (p. 61)

Figure 23: GtC view from the front entrance (p. 62–63)

Figure 24: (not labeled in text; appears as GtC image on p. 63–64)

Figure 25: Patch of a variety of tomatoes being grown atop a pile of mulch (p. 65)

Figure 26: Author's identification of possible tomato varieties (p. 65–66)

Figure 27: Freshly prepared garlic bed (p. 66)

Figure 28: Garden bed with herbs (p. 67)

Figure 29: Active rows of cauliflower in the left side of garden (p. 68)

Figure 30: Side view of the plastic covered hoop house & adjacent solar panel at the back GtC entrance (p. 69)

Figure 31: Hancock Street Farm shed (p. 70)

Figure 32: Array of seating at GtC's front entrance (p. 71)

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